Performance and Robustness Benchmarking of Battery SOC Estimation Methods for Embedded Applications

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Abstract

Robustness of battery State-of-Charge (SOC) estimation cannot be inferred from nominal accuracy alone. This study compares representatives of four major SOC-estimation classes under one causal and cell-disjoint protocol: fixed-capacity Coulomb counting (DM), SOH-adaptive Coulomb counting (HDM), a second-order equivalent-circuit observer (HECM), and a recurrent data-driven estimator (DD). LFP aging trajectories span multiple load, thermal, and SOH conditions. Performance is separated into nominal accuracy against the dataset ground truth, robustness when measurements are corrupted or unavailable, and recovery after an initialization mismatch. The six-cell macro MAE under nominal inputs is 0.0690 for DM, 0.0412 for HDM, 0.0337 for HECM, and 0.0258 for DD. DD also leads several robustness cases, but no representative dominates every disturbance family and the normalized robustness ordering changes with the aggregation rule. Under the tested model-specific initialization interventions, HECM has the smallest observed persistent-recovery-or-censor time at 1.20 h, compared with 2.60 h for DD. Earlier returns cannot be resolved during the common DD warm-up interval. The cell-macro persistent-censor fraction is 0.89 for both DM and HDM. Aggregated scores are therefore retained only as illustrative decision aids rather than a universal ranking. The isolated SOC

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cores reproduce their software references and meet the tested one-second STM32 deadline, while direct latency, flash, and runtime-RAM measurements expose substantial implementation trade-offs. The results demonstrate why estimator selection must combine accuracy, disturbance-specific robustness, recovery, and embedded feasibility. Conclusions apply to the tested representatives and LFP data rather than their complete model families.

Keywords: Battery management system, State of charge, Robustness benchmark, Fault tolerance, Equivalent circuit model, Recurrent neural network, Embedded implementation, Microcontroller

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Nomenclature

Acronyms

BMS	Battery management system
SOC	State of charge
SOH	State of health
DM	Direct-measurement model
HDM	Hybrid direct-measurement model
HECM	Hybrid equivalent-circuit model
DD	Data-driven neural model
ECM	Equivalent circuit model
EKF	Extended Kalman filter
MAE	Mean absolute error
RMSE	Root-mean-square error
ADC	Analog-to-digital converter
CV	Constant-voltage phase
OCV	Open-circuit voltage
GRU	Gated recurrent unit
LSTM	Long short-term memory
LFP	Lithium iron phosphate

Symbols

t	Time
$I(t)$	Current signal at time t
$U(t)$	Voltage signal at time t
$T(t)$	Temperature signal at time t
$\widehat{\mathrm{SOC}}(t)$	Estimated state of charge at time t
$\widehat{\mathrm{SOH}}(t)$	Estimated state of health at time t
$\mathrm{SOC}_{\mathrm{ref}}(t)$	Dataset SOC ground truth at time t
$\mathrm{SOH}_{\mathrm{ref}}(t)$	Reference SOH target at time t
b_I	Multiplicative current-gain error
ΔU	Voltage bias or offset
ΔT	Temperature bias or offset
σ_I	Standard deviation of injected current noise
σ_U	Standard deviation of injected voltage noise
σ_T	Standard deviation of injected temperature noise
Q_c	Online cumulative charge state used by the SOC models
C_{nom}	Nominal capacity
C_{eff}	Effective capacity used inside the estimator
$U_{\max}$	Upper voltage threshold used for full-charge detection
U_{tol}	Voltage tolerance used in the full-charge condition
U_1, U_2	Polarization voltages of the first and second RC branch
$\Delta\mathrm{MAE}$	Error increase relative to baseline

1. Introduction

1.1. Motivation and practical relevance

Reliable battery state estimation is a core BMS function because SOC and SOH determine safety margins, usable energy, operating limits, charge control, and degradation-aware system management in both mobile and stationary lithium-ion battery applications [1, 2, 3]. Practical deployment quality is not defined by clean-condition accuracy alone, because real BMS operation is exposed to sensor gain and offset errors, additive noise, initialization uncertainty after restart, missing samples, irregular sampling, communication interruptions, and transient signal spikes caused by measurement-chain limitations or disturbances in the surrounding system [4, 5, 6]. The relevant engineering question is therefore not only whether an estimator achieves low MAE under nominal measurements, but also whether it converges back to a reliable state after disturbed initialization and whether it remains stable under physically plausible measurement corruption.

As summarized conceptually in Figure 1, the present benchmark treats deployment-facing estimator quality as a three-dimensional problem consisting of nominal accuracy, recovery after state inconsistency, and robustness against disturbed measurements and signal-integrity faults, embedded within broader constraints on memory and execution time. The central question is how four structurally different SOC-estimation classes, represented by the implementations studied here, respond to common failure mechanisms and microcontroller constraints, and whether their nominal ranking remains informative under those conditions. This distinction is essential because one implementation can be accurate under clean measurements yet difficult to re-anchor, sensitive to a specific input fault, or too costly under its trained inference semantics.

1.2. State of the art in battery state estimation

The literature commonly groups battery-state estimators into direct-measurement or integration-based approaches, model-based approaches based on equivalent-circuit or electrochemical models, hybrid approaches that combine physical structure with learned components, and purely data-driven models based on neural sequence learning [7, 1, 2, 3]. Direct-measurement approaches, typically based on Coulomb counting, are attractive because they are simple, computationally inexpensive, and easy to deploy. However, they remain structurally sensitive to current-gain error, missing data, and initialization errors because

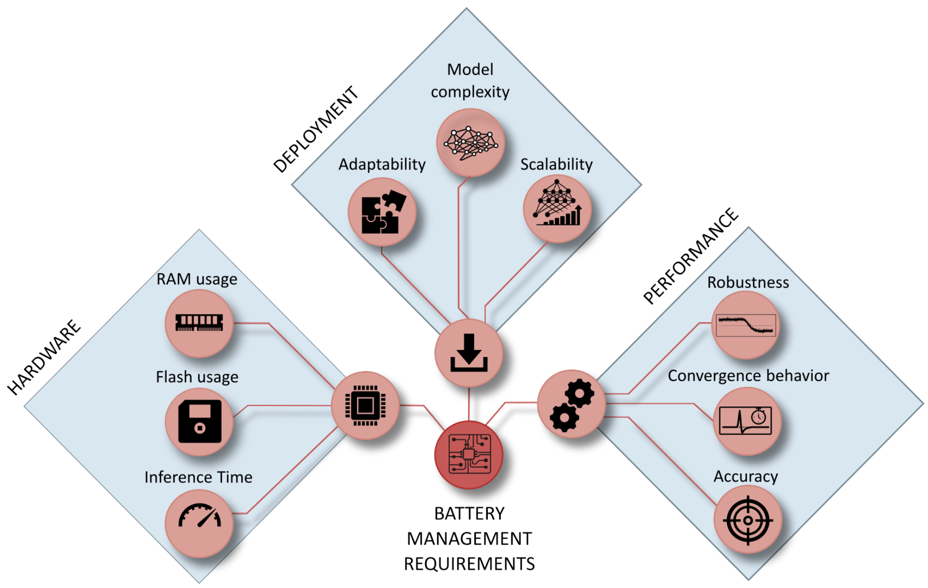

Figure 1: Conceptual overview of deployment-oriented requirements for embedded BMS state estimation. The broader requirement space includes implementation effort, hardware constraints, and estimator performance, whereas the present study focuses on the performance branch and evaluates baseline accuracy, convergence behavior, and robustness under measurement disturbances.

they lack an internal correction mechanism [7, 4]. Model-based approaches, especially ECM- and observer-based estimators, offer physical interpretability and voltage-based correction capability, but their practical behavior depends strongly on the quality of model parametrization, operating-point observability, and measurement quality [8, 2, 4].

Hybrid estimators seek to combine physical state propagation with learned adaptation of capacity, parameters, or auxiliary states, which has made joint SOC/SOH estimation an active research direction [9, 10, 11]. This combination can improve baseline calibration, while it does not necessarily eliminate structural sensitivity to measurement disturbances. Data-driven approaches based on recurrent or sequence models can exploit nonlinear cross-relations between current, voltage, temperature, and derived online features, and recent work has also demonstrated their embedded feasibility through TinyML-oriented implementations, pruning, and quantization [12, 13, 14, 15, 3].

1.3. Performance and embedded deployment gap

A large share of the SOC estimation literature still emphasizes clean-data accuracy, average error metrics, or computational efficiency, while comparatively fewer studies analyze how estimators behave under realistic measurement faults and timing corruption [1, 13, 14, 15]. Existing robustness studies often remain confined to a single model family, for example observer-based robustness against modeling and measurement errors [4] or robust estimation concepts for specific BMS architectures under noise and uncertainty [5], whereas cross-family comparisons under the same disturbance protocol and under comparable embedded constraints remain limited. Benchmarking studies in the broader BMS context underline the importance of reproducible scenario design and deployment-relevant validation, but they do not by themselves close the gap between clean-data estimator comparison and measurement-level performance benchmarking across fundamentally different estimator classes [6]. This gap is especially relevant in embedded BMS design, where estimator selection must jointly consider nominal accuracy, convergence, robustness under disturbed measurements, and computational constraints.

1.4. Contribution and study objectives

This work addresses the gap through three linked questions: whether clean-data accuracy predicts performance under measurement and signal-integrity disturbances, which structural mechanisms govern degradation and recovery, and how the resulting behavior trades against embedded execution cost. To answer them, four deployment-oriented representatives are executed under one causal protocol on cell-disjoint LFP aging data covering multiple load and SOH states. The campaign combines physically interpretable measurement, timing, availability, quantization, and initialization disturbances with a common recovery criterion, cell-level uncertainty analysis, a paired SOH-input ablation, and STM32 measurements of the isolated SOC cores.

Table 1: Estimator-class envelope and scope of the four benchmark representatives. Results apply to the selected implementations in the last column, not automatically to every method listed in the broader class envelope.

Class	Minimum defining mechanism	Broader class can include	Representative used in this study
Direct measurement	Causal integration of measured current using an assumed capacity	Coulombic efficiency, capacity adaptation, OCV or end-point anchoring, plausibility logic, and reset thresholds	Fixed-capacity Coulomb counting with a sustained full-charge CV reset and no voltage-residual correction
Hybrid direct measurement	Integration core augmented by an estimated or learned state/parameter	Learned capacity or efficiency, residual correction, multi-sensor fusion, and adaptive anchoring	Same Coulomb-counting and reset logic as DM, with effective capacity supplied by the shared causal LSTM-SOH estimate
ECM observer	Coulomb-counted SOC propagation plus a voltage model and feedback correction	Higher RC order, hysteresis, thermal states, SOC/SOH/temperature maps, and different nonlinear observers	Second-order RC model with EKF correction, charge/discharge SOC–SOH lookup surfaces, and shared causal SOH input
Data driven	Learned mapping from measured signals or their history to SOC	Feed-forward, convolutional, recurrent, attention-based, physics-informed, raw-signal, or engineered-feature models	Structurally pruned rolling-window GRU using voltage, current, temperature, SOH, Q_c, derivatives, and Δt

The resulting evidence supports comparisons among these four implementations under the declared protocol. It does not establish a universal ranking of direct-measurement, hybrid, observer-based, or data-driven families, whose behavior can change with anchoring logic, parameterization, architecture, features, training data, and deployment semantics.

2. Methodology

2.1. Estimator representatives and class boundaries

Figure 2 summarizes the causal benchmark chain, from measured signals and online feature reconstruction to disturbance injection and global and local evaluation. The four estimators are selected to expose distinct correction mechanisms, not to span every implementation possible within each class. Because class boundaries overlap, each representative is assigned according to its dominant SOC update and correction mechanism. Table 1 separates this defining principle from optional extensions and from the exact implementation evaluated here. There is no meaningful universal “maximum” model within a class: adding learned corrections, higher-order dynamics, richer sensing, or more complex sequence architectures progressively changes both capability and deployment cost.

All SOH-dependent representatives, namely HDM, HECM, and DD, receive the same causal LSTM-SOH trace so that differences arise from how the SOC branch consumes that context. DM is the low-complexity integration reference.

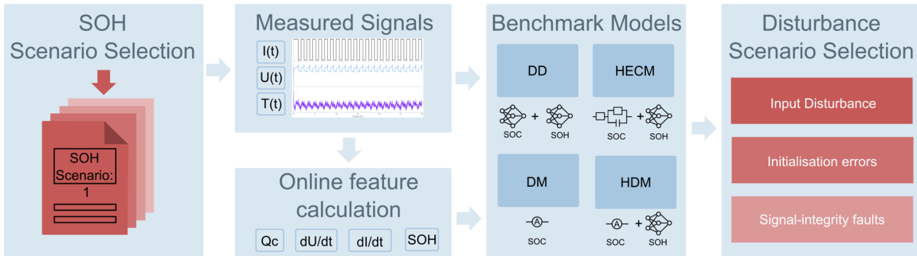

Figure 2: Methodology overview for the causal benchmark chain. Measured battery signals are converted into online auxiliary features, processed by representative estimator implementations, exposed to measurement and initialization interventions, and finally evaluated through global and local performance metrics.

HDM isolates the effect of learned capacity adaptation without adding voltage feedback. HECM adds model-based voltage-residual correction to current-driven state propagation. DD represents compact temporal learning. The GRU was selected because SOC and disturbance recovery depend on recent signal history while its gated single-state structure is lighter than a comparable LSTM. Because DD also consumes the engineered Q_c feature, it is a data-driven sequence estimator with physics-informed inputs rather than a raw-signal-only neural model.

2.2. SOC estimation methods

2.2.1. Direct-measurement model (DM)

The direct-measurement SOC formulation used in DM, HDM, and as an explicit auxiliary feature in DD follows the same online cumulative-charge state, denoted here uniformly as Q_c. It is updated causally from the measured current stream as

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600\,\mathrm{s\,h^{-1}}}, \tag{1}$$

where I_k is given in ampere and Δt_k in seconds, so that the factor $3600\,\mathrm{s\,h^{-1}}$ converts the increment to ampere-hours. The sign convention is fixed by the benchmark implementation and the state is reset to $Q_c = 0$ after a sustained constant-voltage full-charge event. In the DM, this same charge state is mapped directly to SOC through the fixed nominal capacity

$$\widehat{\mathrm{SOC}}_{\mathrm{DM}}(k) = \mathrm{clip}\left(1 + \frac{Q_c(k)}{C_{\mathrm{nom}}}, 0, 1\right). \tag{2}$$

2.2.2. Hybrid direct-measurement model (HDM)

The HDM preserves exactly the same Coulomb-counting core and differs only in the capacity term. The SOH estimate rescales the usable capacity as

$$C_{\mathrm{eff}}(k) = C_{\mathrm{nom}} \cdot \widehat{\mathrm{SOH}}(k), \tag{3}$$

which yields

$$\widehat{\mathrm{SOC}}_{\mathrm{HDM}}(k) = \mathrm{clip}\left(1 + \frac{Q_c(k)}{C_{\mathrm{eff}}(k)}, 0, 1\right). \tag{4}$$

This formulation makes the relation between DM and HDM explicit: both use the same online charge state Q_c, while only the denominator changes through online SOH support. In both direct-measurement variants, a sustained near-maximum voltage condition is interpreted as a constant-voltage full-charge event. If the voltage remains above $U_{\max} - U_{\mathrm{tol}}$, where U_{tol} denotes the admissible voltage margin below the upper threshold $U_{\max}$, for the configured CV duration, the internal charge state is reset so that $Q_c = 0$ and therefore $\widehat{\mathrm{SOC}} = 1$. This reflects the practical assumption that a completed CV phase provides a physically anchored full-charge reference.

2.2.3. Hybrid equivalent-circuit model (HECM)

The hybrid ECM model is implemented as a second-order RC model with the state vector

$$\mathbf{x}_k = \begin{bmatrix} \widehat{\mathrm{SOC}}_k & U_{1,k} & U_{2,k} \end{bmatrix}^{\top}, \tag{5}$$

where U_1 and U_2 denote the polarization voltages of the two RC branches. The prediction step propagates the state according to

$$\mathbf{x}_k^- = \mathbf{A}_k \mathbf{x}_{k-1} + \mathbf{b}_k I_{k-1}, \tag{6}$$

with

$$\mathbf{A}_k = \mathrm{diag}\left(1, e^{-\Delta t_k/\tau_1}, e^{-\Delta t_k/\tau_2}\right), \quad \mathbf{b}_k = \begin{bmatrix} \eta \Delta t_k / C_b \\ R_1\left(1 - e^{-\Delta t_k/\tau_1}\right) \\ R_2\left(1 - e^{-\Delta t_k/\tau_2}\right) \end{bmatrix}, \tag{7}$$

where $C_b = C_0\widehat{\mathrm{SOH}}$ is the SOH-scaled available capacity. The corrected terminal-voltage estimate is written as

$$\widehat{U}_k = \mathrm{OCV}(\widehat{\mathrm{SOC}}_k, \widehat{\mathrm{SOH}}_k, I_k) + U_{1,k} + U_{2,k} + R_i I_k, \tag{8}$$

and the Kalman update uses the residual between $\widehat{U}_k$ and the measured terminal voltage to correct the internal state. In the present implementation, the state propagation is driven by the previous current sample I_{k-1}, whereas the measurement equation uses the current sample I_k available at the correction step. The ECM parameters are not constant. The implementation interpolates R_i, R_1, R_2, τ_1, τ_2, OCV, and dOCV/dSOC from a preidentified lookup table over SOC and SOH, with separate charge and discharge parameter surfaces. The HECM is implemented as a two-RC observer with SOH-dependent parameter scheduling.

2.2.4. Data-driven SOC model (DD)

The data-driven SOC model is a GRU-based recurrent estimator with the feature vector $\{U$ [V], I [A], T [°C], SOH, Q_c, dU/dt, dI/dt, $\Delta t\}$. The benchmarked, structurally pruned and fine-tuned model consists of a single-layer GRU with hidden size 67 followed by an MLP head with one hidden layer of size 96 and a sigmoid output layer. During optimization, the recurrent core is exposed to causal sequence segments of length $L_{\mathrm{SOC}} = 2024$ samples. Within this feature set, Q_c is exactly the same online cumulative-charge state defined in (1). It is included explicitly so that the neural estimator receives the same charge-throughput information that also drives DM and HDM. The derivative channels are computed online as the finite differences $\mathrm{d}I/\mathrm{d}t(k) = (I_k - I_{k-1})/\Delta t_k$ and $\mathrm{d}U/\mathrm{d}t(k) = (U_k - U_{k-1})/\Delta t_k$. They provide local transient information that is not captured by the absolute channels alone. The additional feature Δt_k allows the network to distinguish amplitude changes from sampling-interval changes under irregular timing. During the primary benchmark, each output is calculated from the latest causal 2024-sample window and recurrent state is reset between windows, matching training and validation. No future information is used. Continuous hidden-state forwarding is evaluated only as a separate deployment ablation because it changes the learned inference semantics.

2.3. SOH estimation method

The shared SOH estimator used in the current benchmark is an LSTM-based sequence-to-sequence model that operates on hourly aggregated online features derived from $\{U \text{ [V]}, I \text{ [A]}, T \text{ [°C]}, \text{EFC}, Q_c\}$, where the aggregation uses the statistics mean, standard deviation, minimum, and maximum for each feature [16, 17, 18, 19]. This transforms the five base channels into a 20-dimensional hourly feature vector. Architecturally, the SOH network first projects the aggregated input through a two-layer feature embedding block with size 128, layer normalization, GELU activations, and dropout, then processes the sequence with a two-layer unidirectional LSTM of hidden size 112, and finally maps the recurrent output through three residual MLP blocks and a prediction head with hidden size 160 to obtain the SOH estimate for each hourly step.

During optimization, the SOH estimator is exposed to causal hourly sequence segments, and benchmark execution processes one hourly aggregate at a time with causal hidden- and cell-state forwarding. This allows the estimator to accumulate degradation context without access to future information. The first hourly output is anchored with the configured initial SOH value before subsequent hourly predictions are taken directly from the recurrent model output. The SOH prediction is updated at hourly cadence and then held piecewise constant between update points, which enforces an online deployment constraint and avoids sample-wise access to future information while still allowing the faster SOC estimators to use the latest SOH context. This cadence matches the slowly varying capacity fade observed in the dataset but cannot represent an abrupt capacity-loss event, which is absent from the measurements and outside the study claims. In the HDM, the predicted SOH enters the SOC computation through the effective capacity relation $C_{\text{eff}} = C_{\text{nom}} \cdot \widehat{\text{SOH}}$, so the method remains structurally close to direct measurement while replacing the fixed-capacity assumption. In the HECM, the predicted SOH affects the capacity scaling and the lookup-based ECM parametrization, while in the DD model it is provided as an explicit SOC input feature that contextualizes the current measurement trajectory.

2.4. Model and target-hardware preparation

The software benchmark uses trained floating-point model parameters and input scalers fixed before holdout evaluation. Neural training, validation, hyperparameter selection, scaler fitting, pruning, and final model selection use only the declared training and validation cells. No holdout cell enters these steps. The HECM lookup surfaces are HPPC-derived from cells assigned to the training and validation development pool. They exclude the six holdout cells and were fixed without retuning before benchmark evaluation. The compact SOC-GRU has hidden size 67 and the shared SOH-LSTM has hidden size 112. All neural configurations, parameter sets, and preprocessing definitions were fixed before holdout evaluation. The evaluation procedure accepts only cells from the declared holdout set unless an analysis is explicitly identified as diagnostic.

The target-hardware experiment evaluates the isolated SOC cores on a NUCLEO-H753ZI board with a 480 MHz STM32H753 Cortex-M7. To separate SOC implementation cost from the shared SOH service, the board receives the precomputed causal SOH trace used by the software reference. The SOH-LSTM itself is not executed on the microcontroller. Numerical equivalence and inference latency are measured directly on the board. Execution time is recorded for every valid inference with the processor-integrated hardware cycle counter, and each cell sequence is replayed three times. Flash occupancy is calculated from the compiled firmware regions that must be stored in the microcontroller's non-volatile program memory.

Runtime RAM is divided into statically allocated variable storage, dynamically allocated memory, and temporary memory used during nested function calls. The static component includes both initialized variables and variables set to zero during system startup. Maximum temporary function-call memory is measured by filling the available stack region with a known pattern before initialization. After a representative 2025-sample replay of cell C09, the deepest overwritten address identifies the maximum function-call memory used. Dynamic-memory management is instrumented to record the maximum cumulative allocation reached during execution. The reported peak RAM combines these three contributions and therefore covers startup, protocol handling, and estimator execution without counting an unused memory safety reserve configured during compilation as consumed memory. The exact rolling-window DD produces two valid calls after its 2023-sample warm-up in this memory replay, so its measured stack maximum applies to those calls, while its dominant static window allocation is determined at link time. Predictions are compared with both the software implementation and dataset SOC. The experiment characterizes the isolated SOC implementations rather than end-to-end SOC–SOH system behavior.

2.5. Robustness benchmark design

The measurement-disturbance branch manipulates only quantities that a real BMS measures or derives online from measured data, namely current, voltage, temperature, sample availability, and sampling time. A separate initialization branch intervenes in deployment-accessible estimator states. Artificial parameter mismatch and synthetic neural hidden-state corruption are excluded from the primary cross-model ranking [6]. Parameter mismatch is used only in the separately reported HECM lookup-table sensitivity. Figure 3 groups the primary benchmark into input disturbances, initialization errors, and signal-integrity faults, while Table 2 defines the exact disturbance implementations used in the study.

The disturbance magnitudes are not intended as one-to-one copies of a single field specification. Instead, they were selected from three justification classes: deployment-level sensing limitations, adverse measurement corruption, and deliberate signal-integrity stress tests [20, 21, 22, 23, 24, 25]. The main benchmark matrix follows the scenario families shown in Figure 3 and provides the basis of the main result figures and tables. ADC quantization is part of the same

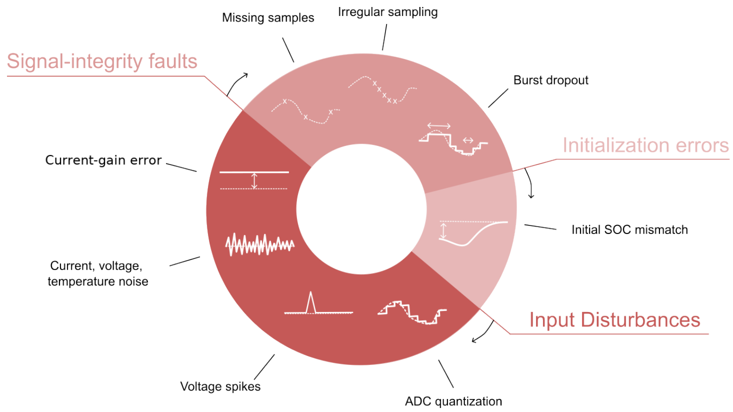

Figure 3: Taxonomy of the disturbance scenarios used in the performance benchmark, grouped into input disturbances, initialization errors, and signal-integrity faults.

scenario set and is reported in Figure 16. Overall, the benchmark is designed as a controlled and reproducible scenario study that exposes estimator-specific failure modes under a common disturbance protocol rather than emulating the complete system stack of a production BMS.

Table 2: Disturbance configurations used in the benchmark and their practical rationale. Exact numeric values are fixed benchmark settings. The cited sources motivate disturbance type and magnitude range.

Scenario	**Benchmark setup**	**Practical origin / rationale**
ADC quantization	$\Delta I = 0.01\,\mathrm{A}$, $\Delta U = 0.005\,\mathrm{V}$, and $\Delta T = 0.5\,^\circ\mathrm{C}$	ADC steps, sensor scaling, rounding, and coarse deployment-level discretization [20, 21, 26]
Current gain error	$I_{\mathrm{meas}} = I(1+b)$ with $b = \pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$. Both signs are evaluated at each magnitude	Sensor gain error, miscalibration, drift, and upper bound as stress case [21, 22, 23]
Current noise	Gaussian with $\sigma_I = 0.02\,\mathrm{A}$ and $0.10\,\mathrm{A}$	Sensing-chain noise, EMI pickup, and low and high sensor-noise levels [4, 5]

Table 2: Disturbance configurations used in the benchmark and practical rationale (continued).

Scenario	**Benchmark setup**	**Practical origin / rationale**
Voltage noise	Gaussian with $\sigma_U = 0.01\,\mathrm{V}$	Wiring noise, front-end noise, poor filtering, and EMI [4, 20, 24]
Temperature noise	Gaussian with $\sigma_T = 1.0\,^{\circ}\mathrm{C}$	Sensor tolerance, conversion noise, thermal gradients, and sensor-to-cell mismatch [26, 11]
Voltage offset	Additive $U_{\mathrm{meas}} = U + 0.02\,\mathrm{V}$	Front-end offset, calibration residual, wiring contribution, and adverse sensing stress
Temperature offset	Additive $T_{\mathrm{meas}} = T + 3\,^{\circ}\mathrm{C}$	Sensor placement error, calibration residual, and cell-to-sensor thermal mismatch
Initial SOC error	Initial mismatch of -0.10 SOC, with DD perturbed through online Q_c	Restart, storage, missing recent history, and weak LFP OCV anchoring [4, 8]
Missing samples	Every 50th sample frozen and separate random 2% loss	Packet loss, blocked updates, logger dropouts, and periodic and random data freeze [5, 25]
Irregular sampling	Uniform jitter: $\pm 0.1\,\mathrm{s}$, $\pm 0.5\,\mathrm{s}$, and $\pm 0.9\,\mathrm{s}$ around nominal $1\,\mathrm{s}$ grid	Scheduling jitter, bus latency, and asynchronous acquisition [6, 5, 25]
Burst dropout	One frozen $3600\,\mathrm{s}$ interval selected by maximum absolute net charge, with at least 12 h before and 24 h after the interval	Communication interruption, frozen task, stuck data stream, and measurement-only severe case [6, 5]
Voltage spikes	Single-sample spikes of $\pm 0.20\,\mathrm{V}$ every 1000 samples ($1000\,\mathrm{s}$)	EMI, switching disturbance, and corrupted voltage samples [24]

2.6. Metrics and evaluation criteria

For deployment-oriented interpretation, the benchmark results are synthesized along three performance dimensions: nominal accuracy under baseline conditions, recovery after initialization mismatch, and robustness under sustained or transient measurement corruption and signal unavailability. Disturbance-related global performance is quantified through MAE, RMSE, bias, maximum error, and the 95th percentile absolute error, because these metrics provide a compact view of average performance, systematic deviation, and tail behavior across the selected evaluation windows. Local disturbance performance is analyzed through scenario-specific metrics such as jump counts, output variance, absolute-error variance, and excess rolling MAE, because several disturbances do not primarily manifest as window-wide MAE collapse but as short-term output fluctuations, delayed drift, or slow re-synchronization. Whenever a disturbance mask exists, the analysis further separates pre-disturbance, disturbed, and post-disturbance behavior in order to quantify disturbance-window error growth and residual post-disturbance error.

ADC quantization, current-gain error, voltage and temperature disturbances, missing samples, and irregular sampling are interpreted primarily through global metrics. Burst dropout, voltage spikes, current-noise detail, and initialization mismatch additionally use dedicated local evidence because their most relevant effects are episodic. The canonical initialization-recovery analysis compares each perturbed estimator trajectory with its cell-, window-, model-, SOH-, and seed-matched correctly initialized trajectory. A first qualifying entry occurs when the absolute paired difference enters a 0.02 SOC band for at least 300 s. Persistent recovery is the first subsequent point after which the difference remains within this band for the rest of the 24-h observation horizon. Persistent recovery is the primary time endpoint. Runs without this endpoint are right-censored, while first-entry time and a later departure from the band are retained as diagnostics. The comparison reports both times, censored and relapse fractions, and excess-error area under the curve. It never substitutes absolute estimator error against dataset SOC for the paired recovery difference. The requested -0.10

initialization mismatch is mapped into each estimator's available online state: the coulomb-counting SOC initialization for DM and HDM, the EKF SOC state for HECM, and the capacity-equivalent Q_c offset for DD. Because the DD mapping passes through the recurrent network rather than directly fixing its output, the difference at the first commonly scored sample is recorded before recovery is interpreted. The common observation starts at source sample 2023, approximately 0.562 h after the intervention. An endpoint already satisfied at this first observation is left-censored, and the recorded 0.562 h value is a conservative upper bound rather than an exact recovery time. The resulting comparison concerns deployment-accessible initialization interventions, not mathematically identical latent-state perturbations.

The independent statistical unit is the holdout cell, not an individual time sample or selected window. Protocol-defined SOH windows are first averaged within each cell and random seed. The six cells then receive equal macro weight. Hierarchical bootstrap resampling with seeds nested inside cells and 10,000 repetitions provides 95% confidence intervals. Scenario effects are evaluated as paired, cell-matched differences from baseline, and estimator pairs are compared using exact two-sided sign-flip tests, raw mean and median differences, paired standardized effect sizes, and Holm-adjusted p-values. Holm correction is applied within each reported scenario family. Because only six independent cells are available, effect magnitudes and confidence intervals are treated as primary evidence and p-values as supporting evidence. The full machine-readable tests accompany the result tables, while the baseline pair tests are reproduced in the Appendix.

The rolling DD implementation first emits an SOC estimate at source sample 2023 because it requires a 2024-sample causal input window. All global accuracy, robustness, stratified, confidence-interval, and paired-test calculations therefore use the common source-sample mask $k \geq 2023$ for every estimator. Each 24-h window contributes 84,377 matched evaluation points per estimator, while each 48-h dropout window contributes 170,777. This prevents the other estimators from receiving an additional early interval that DD cannot evaluate.

The decision synthesis is explicitly illustrative. Accuracy combines baseline MAE, RMSE, and 95th-percentile error. Robustness assigns equal weight to eight evaluated disturbance families: sensor noise, current-gain error, sensor offsets, quantization, missing samples, timing jitter, burst dropout, and voltage spikes. The current-gain family uses the adverse direction from the matched signed sweep in the final benchmark build. Non-positive ΔMAE values are treated as zero disturbance penalty. Levels within a family are averaged before the family scores are combined, so families with more tested magnitudes do not receive extra weight. Recovery combines three outputs from the single canonical paired initialization analysis: observed persistent recovery-or-censor time, excess-error area, and persistent censored fraction. Relapse after a first qualifying entry remains a separate diagnostic because it is undefined for runs that never enter the band and must not reward non-recovery. Alternative equal-scenario and high-severity weighting variants are retained as sensitivity analyses. The normalized synthesis is a relative comparison within these four representatives and is not treated as an objective estimator-family ranking.

3. Experimental Setup

3.1. Dataset and data acquisition

The benchmark uses the MGFarm LFP 18650 dataset because its controlled design-of-experiments aging study provides long 1-Hz full-cell trajectories, repeated capacity checkups, multiple operating conditions, and enough independent cells for a cell-disjoint comparison [27, 28, 29]. Model development and final testing use a fixed split. SOC and SOH training use C01, C03, C05, C11, C17, and C23. Validation uses C07, C19, and C21. The independent holdout benchmark uses C09, C13, C15, C25, C27, and C29. The six test cells span distinct aging, thermal, duration, and load envelopes, as summarized in Figure A.21. LFP's flat mid-SOC OCV and chemistry-specific aging behavior are integral to the tested problem.

The holdout cells are assigned to descriptive load groups using only the measured 95th-percentile absolute C-rate and no estimator output or error: C25 and C27 form the low-load group, C09, C13, and C15 the middle-load group, and C29 the high-load case. Because the high-load group contains one cell, its group-by-SOH results are explicitly exploratory and are not interpreted as population-level high-load inference. Within each trajectory, performance is additionally stratified by reference SOH (fresh at SOH ≥ 0.90, mid-life at $0.80 \leq$ SOH < 0.90, and aged at SOH < 0.80), instantaneous absolute C-rate (< 0.5, 0.5–1.5, and ≥ 1.5), temperature ($\leq 30\,^{\circ}$C, 30–35 °C, and $> 35\,^{\circ}$C), and SOC (< 0.20, 0.20–0.80, and > 0.80). Absent states are reported as not available and are never imputed.

3.2. Dataset ground truth and preprocessing

The reference preprocessing first computes the signed transferred charge as $Q_k = I_k \Delta t_k / 3600$ and the absolute transferred charge as $|Q_k|$, from which the capacity during dedicated capacity-test intervals is obtained as $C_{\text{ref}} = \sum_k |Q_k|$. The reference SOH target is then defined offline as $\text{SOH}_{\text{ref}} = C_{\text{ref}} / C_{\text{ref},0}$, where $C_{\text{ref},0}$ denotes the first measured reference capacity of the trajectory, and the resulting capacity and SOH columns are interpolated between the dedicated capacity-test anchors. The dataset SOC ground truth used in the benchmark is reconstructed offline as $\text{SOC}_{\text{ref}} = 1 + Q_c / C_{\text{ref},0}$, where the cumulative charge state Q_c is propagated from the signed charge and reset to 0 at the upper voltage anchor and to $-C_{\text{ref},0}$ at the lower voltage anchor during preprocessing. This reconstructed SOC serves as the common dataset ground truth for every reported SOC error. It is an offline evaluation quantity because its dataset-side processing and anchor logic are not fully available to an online estimator during deployment. All nominal accuracy values therefore refer to this consistently applied dataset ground truth. Paired ΔMAE compares each disturbed run with its matched baseline under the same ground truth.

3.3. Test trajectory and benchmark scenarios

All estimators are evaluated on every available holdout-cell/SOH combination under the same disturbance protocol. One representative 24-h window is frozen for each available fresh, mid-life, and aged state, yielding 16 windows. C27 contributes only a fresh window because its measured SOH does not fall below 0.90. Windows start at a measured full-charge anchor (SOC $\geq$ 0.98, $U \geq 3.58\,\mathrm{V}$) and are selected as measured-feature medoids without using estimator outputs or errors. The one-hour dropout uses 48 h from the same anchor to retain approximately 24 h of post-gap observation. The shared SOH LSTM receives 192 h of undisturbed causal context before every scored window, matching its trained sequence length. Figure A.23 summarizes this protocol.

The final benchmark build contains 19 scenario definitions: the nominal baseline, two current-noise levels, voltage and temperature noise, three signed current-gain magnitudes, voltage and temperature offsets, ADC quantization, initial SOC mismatch, periodic and random sample loss, three timing-jitter levels, a one-hour burst dropout, and randomized-sign voltage spikes. Each current-gain magnitude contains matched positive and negative sign sublevels. Adverse-direction summaries report the larger paired ΔMAE of the two signs at each magnitude and thereby prevent signed compensation with baseline error from being interpreted as robustness. Gaussian sensor-noise cases use 10 deterministic seeds. Random sample loss, jitter, and spike cases use 5 seeds. Deterministic cases run once. Figure A.22 maps the final scenario definitions to the manipulated measurement or state and marks the selected paired reference-SOH ablations. Input disturbances are applied directly to measured channels, initialization interventions are mapped to each model's deployment-accessible state with their non-equivalence disclosed, and signal-integrity scenarios alter sample availability or timing while preserving causal estimator execution.

3.4. Implementation details

Every estimator is executed causally, and all auxiliary quantities required by the models are rebuilt directly from disturbed inputs. This applies to Q_c, EFC,

dU/dt, dI/dt, and hourly SOH aggregates, so no hidden dataset-side feature leaks into the estimator input. The SOH LSTM forwards hidden and cell states across hourly updates after its 192-h initialization context. The primary SOC-GRU evaluation preserves its trained sequence-to-one semantics: each prediction uses the latest 2024-sample rolling window with recurrent state reset between windows. A faster continuous-stateful stream is treated only as a deployment ablation because it changed full-trajectory DD accuracy and therefore cannot silently replace the trained benchmark semantics. In freeze and dropout scenarios, disturbed or frozen measurements propagate consistently into derived online features. The dropout start is selected from measured current only as the eligible one-hour interval with the largest absolute integrated net charge. Selection enforces at least 12 h of pre-gap observations and 24 h of post-gap observations and never uses dataset ground-truth SOC or estimator output. Direct-measurement estimators use the same CV-triggered reset that defines the full-charge start, whereas the DD initial-SOC test uses an equivalent offset of online Q_c because the network exposes no explicit SOC state. The environment enforces common nominal-capacity metadata, disturbed streams, targets, masks, and metrics. Model-specific capacity handling remains part of the estimator definitions.

3.5. HECM lookup-table sensitivity

A separate HECM-only analysis quantifies whether its current-gain response depends on exact lookup-table calibration. It applies one systematic perturbation at a time to the fixed charge and discharge surfaces. The R_i, R_1, and R_2 surfaces are jointly scaled by -10% or $+10\%$, while the OCV surfaces are shifted by $-10\,\mathrm{mV}$ or $+10\,\mathrm{mV}$. The OCV derivative and time-constant surfaces remain fixed. Each of these four perturbations and the nominal lookup are evaluated under baseline current and matched -3% and $+3\%$ current-gain errors over all 16 frozen windows, yielding 240 HECM runs. The analysis uses the scenario-matched causal SOH traces and common source-sample mask. Windows are averaged within each cell before equal-weight six-cell aggregation and a 10,000-

draw cell bootstrap. For each lookup condition, the adverse gain penalty is the larger cell-level ΔMAE from the two gain signs. The lookup–gain interaction subtracts the nominal-lookup adverse penalty from the perturbed-lookup adverse penalty. These one-at-a-time shifts are controlled local sensitivity probes rather than probabilistic uncertainty bounds. They remain outside the cross-model robustness score.

4. Results

4.1. Baseline performance

Figure 4 summarizes nominal accuracy against the dataset ground truth across the 16 frozen windows on the common source-sample interval. Equal weighting of the six holdout cells yields a different ranking from the previous single-trajectory analysis: DD has the lowest baseline MAE at 0.0258 (95% CI 0.0189–0.0325), followed by HECM at 0.0337 (0.0246–0.0434), HDM at 0.0412 (0.0301–0.0515), and DM at 0.0690 (0.0499–0.0823). The RMSE ranking is identical, with values of 0.0300, 0.0388, 0.0442, and 0.0740, respectively. Cell-level points reveal substantial between-cell variation, particularly for DM, so the bars must be interpreted as cell-macro summaries rather than performance on a homogeneous trajectory. Exact paired sign-flip tests support the magnitude and direction of the principal contrasts, but none of the six baseline pair tests remains below 0.05 after Holm correction because only six independent cells are available. Confidence intervals and paired effect sizes are therefore the primary evidence, as reported in the Appendix. These values quantify accuracy against the consistently reconstructed dataset ground truth.

4.2. Current-gain sensitivity

Figure 5 summarizes the matched signed current-gain sweep at $\pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$ in the final benchmark build. For each magnitude and holdout cell, the larger MAE change from the two signs is reported because the opposite sign can partially compensate a pre-existing baseline error. The resulting adverse-direction degradation increases monotonically with gain-error magnitude for

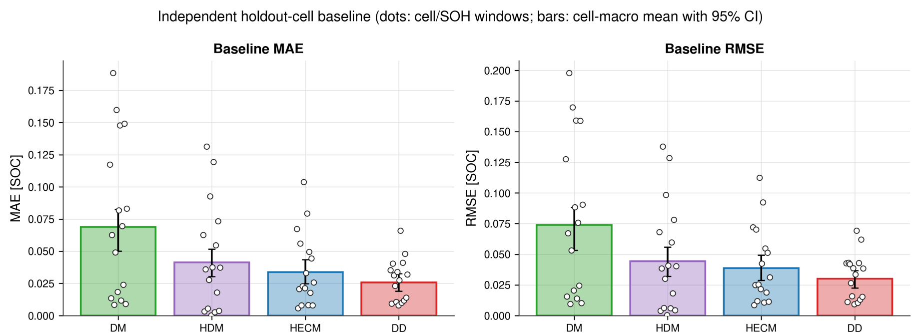

Figure 4: Baseline accuracy against the dataset SOC ground truth across six independent holdout cells. Dots show cell/SOH-window values. Bars and error bars show the equal-weight cell-macro mean and hierarchical 95% confidence interval. DD has the lowest macro MAE and RMSE, while DM has the largest error and between-cell spread.

every estimator. DM and HDM show the strongest response, HECM is less sensitive, and DD is least sensitive. At 3%, the cell-macro MAE increases by 0.0108 for DM, 0.0101 for HDM, 0.0060 for HECM, and 0.0038 for DD. The adverse-direction result is a conservative sensitivity measure and does not imply that the opposite sign is harmless.

The HECM lookup-table analysis in Figure A.25 separates baseline calibration effects from current-gain robustness. The nominal lookup produces a baseline MAE of 0.0337 and an adverse 3% current-gain penalty of 0.00602. Resistance scaling changes baseline MAE to 0.0351 at -10% and 0.0327 at $+10\%$, while the corresponding gain penalties remain 0.00614 and 0.00610. OCV shifts change baseline MAE to 0.0314 at $-10\,\mathrm{mV}$ and 0.0373 at $+10\,\mathrm{mV}$, while their gain penalties are 0.00628 and 0.00580. Relative to the nominal lookup, the four macro interaction effects range only from -0.00022 to $+0.00026$ MAE, and every 95% interval includes zero. Within these local perturbation ranges, lookup calibration affects absolute HECM accuracy more than its incremental current-gain penalty.

The continuous C29 analysis in Figure 6 complements the window-based benchmark and separates three mechanisms that independent 24-h windows

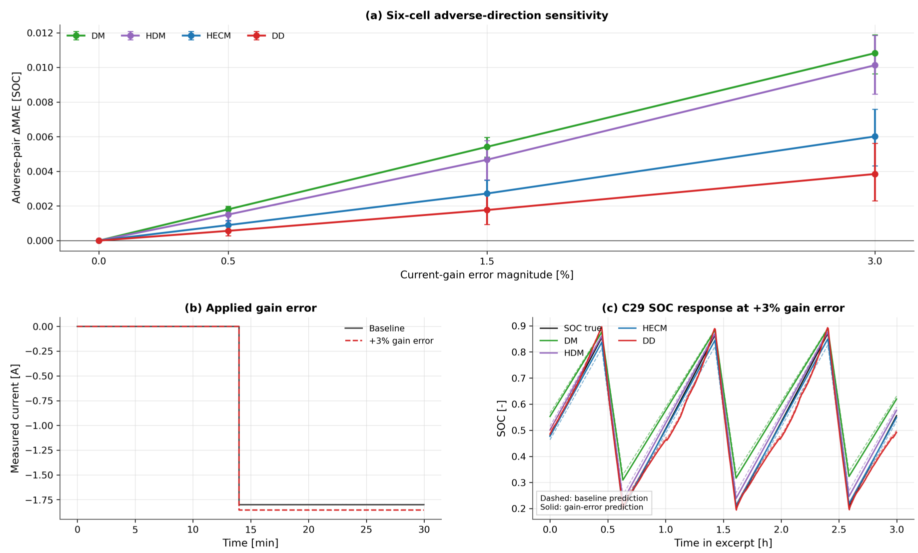

Figure 5: Current-gain sensitivity in the final benchmark build. (a) Six-cell adverse-direction ΔMAE for each matched $\pm b$ pair with hierarchical 95% confidence intervals. (b) Representative current before and after the +3% gain error. (c) C29 SOC trajectories over three charge–discharge intervals. The adverse-pair summary prevents signed compensation of pre-existing baseline bias from being misread as improved robustness.

cannot resolve. The gain-error contribution varies over life because its signed Coulomb-counting error interacts with charge direction and operating state. Between full charges, the penalty often grows within an interval, but full-charge anchoring can partially reduce or restructure it. The process is therefore neither an unrestricted monotonic drift nor a complete reset. Across this single full-life replay, the adverse 3% gain error increases MAE by 0.0102 for DM, 0.0072 for HDM, 0.0037 for HECM, and 0.0048 for DD. HECM therefore has the smallest full-life gain-error contribution in C29, while DM and HDM remain most exposed. This diagnostic is not a six-cell lifecycle ranking.

4.3. Noise sensitivity

Repeated sensor noise is weaker than persistent current gain error, but the response depends on channel and estimator structure. At $\sigma_I = 0.10\,\text{A}$, HECM has the largest mean macro penalty ($\Delta\text{MAE} = 0.0038$), although its interval

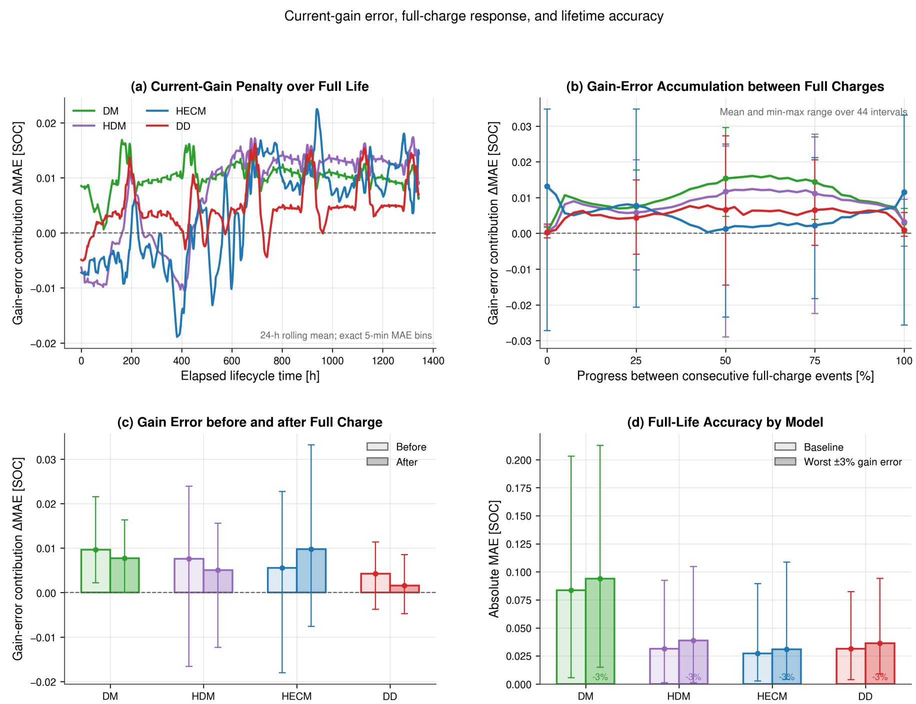

Figure 6: Current-gain-error diagnostic over the continuous C29 life replay. (a) Rolling 24-h gain-error contribution over elapsed life. (b) Mean and range over normalized intervals between consecutive full-charge events. (c) Penalty immediately before and after full-charge events. (d) Full-life baseline and adverse-gain MAE. Full-charge events reduce or restructure accumulated error but do not erase the lifetime penalty uniformly.

spans zero, whereas DM changes by -0.0009 and HDM and DD remain near zero. Small negative changes are paired finite-window effects, not evidence that noise improves an estimator in general. Voltage noise ($\sigma_U = 0.01\,\mathrm{V}$) produces a 0.0014 penalty for DM and 0.0013 for HDM, while HECM and DD remain below 0.0003. Temperature noise is neutral for DM and small for the SOH-aware models. Figure 7 shows the local transmission of current noise, while Figure 8 reports repeated six-cell effects with hierarchical uncertainty.

The offset cases expose a different pattern from zero-mean noise. A $+3\,^{\circ}\mathrm{C}$ temperature offset increases DD MAE by 0.0074 (95% CI 0.0027–0.0125), while the corresponding changes for DM, HDM, and HECM are 0.0000, 0.0002, and

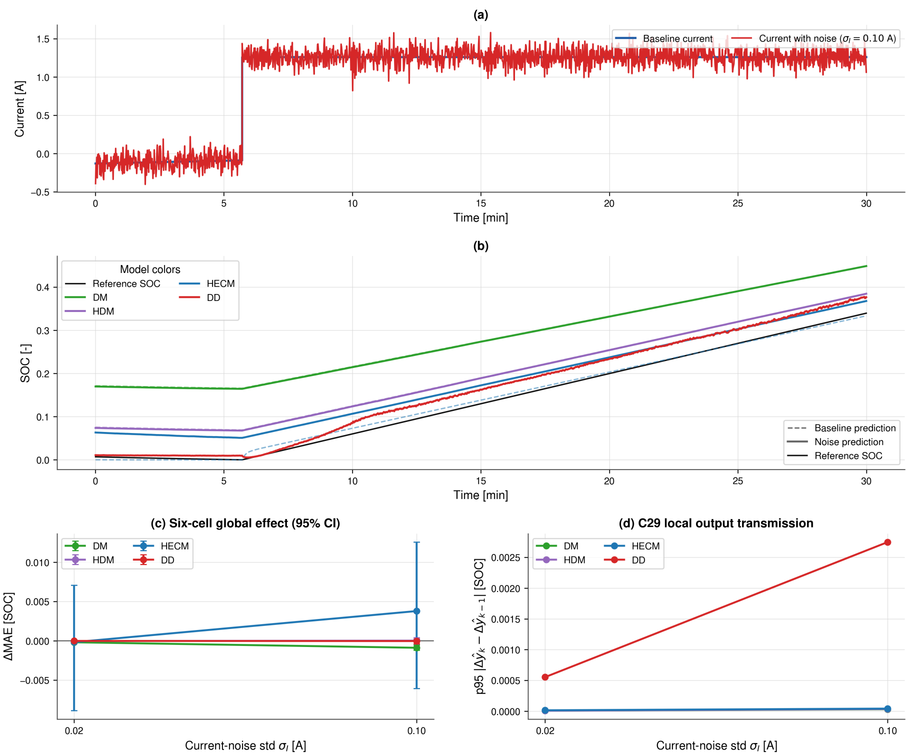

Figure 7: Noise sensitivity under additive current noise. (a) Baseline and noisy current in a reset-free 30 min C29 example. (b) Matched baseline and noisy SOC trajectories in the same window. (c) Cell-macro ΔMAE with hierarchical 95% confidence intervals across six holdout cells. (d) Local C29 transmission measured as the 95th percentile of the change in noise-induced output deviation between consecutive samples. The local panels illustrate the mechanism, while panel (c) provides the population-level comparison.

0.0002. A +0.02 V voltage offset increases DD by 0.0038 and HDM by 0.0005, leaves DM effectively unchanged, and reduces HECM MAE by 0.0040 in these finite windows. The latter signed compensation is not interpreted as a generally beneficial sensor fault. These offset results are included in the cross-scenario heatmap and in the family-balanced robustness synthesis.

4.4. Initialization error and recovery

Initialization performance uses the paired criterion defined above. The same requested −0.10 SOC-equivalent mismatch is mapped to the explicit DM/HDM

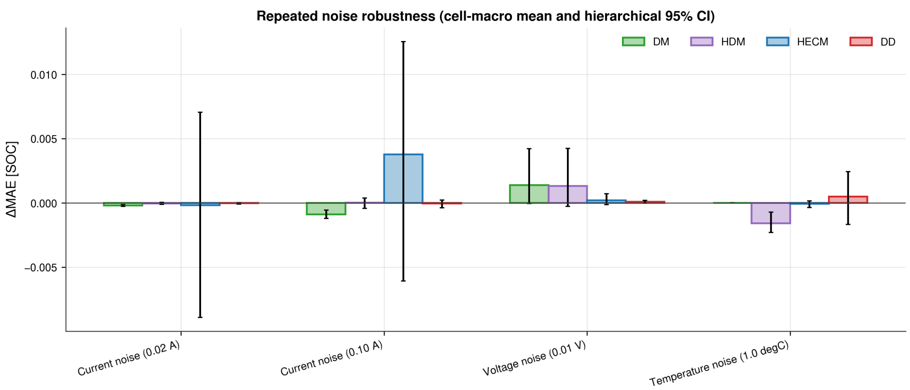

Figure 8: Repeated noise robustness across six holdout cells. Bars show cell-macro ΔMAE and error bars show hierarchical 95% confidence intervals with random seeds nested within cells. HECM is most affected by high current noise, while most other channel–model combinations remain close to baseline relative to the between-cell uncertainty.

Coulomb-counting state, the HECM EKF SOC state, and a Q_c offset of $-0.10C_{\text{nom}}$ for DD. Figure 9 reports the paired trajectory difference rather than assuming that these internal mappings create identical outputs. All four paired trajectories use source samples 2023 onward, while time remains referenced to the intervention at sample 0. The reported observed first-entry and persistent-recovery-or-censor means are 1.12 and 1.20 h for HECM and 1.29 and 2.60 h for DD. HECM and DD already satisfy the persistent endpoint at the first common observation in cell-macro fractions of 0.39 and 0.22, respectively, so those individual times are left-censored and the corresponding means contain conservative upper-bound values. DD's 0.78 cell-macro relapse fraction explains the difference between its first observed entry and persistent return. HECM's corresponding relapse fraction is 0.17. DM first-entry and persistent recovery-or-censor times are both 22.28 h. HDM's first-entry value of 17.21 h increases to 22.28 h under the persistent criterion because temporary entries do not constitute re-anchoring. The cell-macro persistent-censor fraction is 0.89 for both DM and HDM, compared with zero for HECM and DD. HECM also has the smallest excess-error area at 0.078 SOC h, compared with 0.116 SOC h for DD.

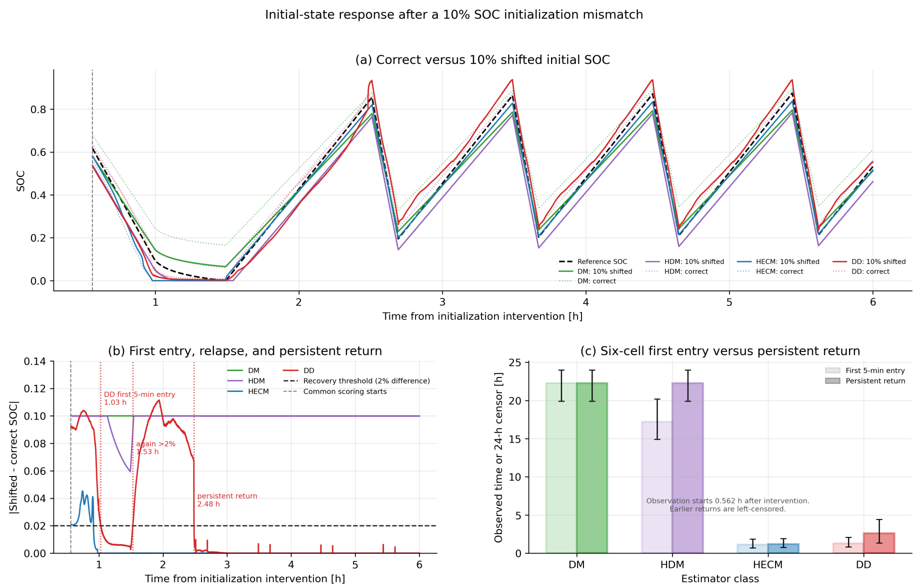

Figure 9: Response to a 10% SOC-equivalent initialization mismatch. (a) Correctly and incorrectly initialized trajectories for the dedicated C29 mid-life pair. (b) Absolute paired differences over the first 6 h. The DD trajectory enters the 2% band for 5 min, leaves it again, and only later returns persistently. The vertical gray marker denotes the first commonly scored sample at 0.562 h. (c) Observed first qualifying entry and persistent recovery-or-censor time across the six holdout cells with hierarchical 95% confidence intervals. Endpoints already satisfied at the common start are left-censored.

Considering the observable paired trajectories, censoring, excess-error area, and relapse together, HECM provides the strongest recovery profile under these model-specific interventions. This does not identify an exact recovery-time distribution during the unobserved warm-up interval. Integration-based states generally require a later physical anchor.

4.5. Missing samples and signal integrity

Figure 10 compares the six-cell effects of periodic and random sample loss with those of timing jitter. Sample loss is substantially more consequential than timing jitter. Periodic freezing of every 50th sample increases MAE by 0.0072, 0.0065, 0.0028, and 0.0011 for DM, HDM, HECM, and DD, respectively.

Random 2% loss produces corresponding increases of 0.0080, 0.0081, 0.0039, and 0.0026. By contrast, even the ± 0.9 s jitter case remains within approximately 6×10^{-4} SOC of baseline for every model, with confidence intervals generally spanning zero. DD therefore has the smallest sample-loss penalty, while DM and HDM retain their structural dependence on uninterrupted charge integration.

The one-hour burst dropout is a distinct availability problem. Against the newly generated duration-matched 48-h baseline, its macro ΔMAE is largest for HECM (0.0239, 95% CI 0.0155–0.0366), followed by HDM (0.0209, 0.0098–0.0313) and DM (0.0050, -0.0029–0.0138). DD changes by -0.0093 (-0.0288–0.0099). The signed DD value indicates finite-window compensation between an already imperfect baseline trajectory and the frozen-input trajectory, not that missing measurements are beneficial. The interval is deliberately placed where the absolute unobserved net charge is largest among starts that retain the required pre- and post-gap context. Figure 11 compares each disturbed C29 output with its duration-matched no-dropout trajectory and reports the six-cell global effect. The paired local deviation has model-specific origins: DM/HDM retain the unobserved charge deficit until a physical anchor, HECM corrects through weak LFP voltage observability, and DD changes as its rolling input context is replaced by valid samples. No six-cell dropout-recovery time is used in the recovery score because recovery is defined only by the canonical paired initialization experiment.

4.6. Spike sensitivity

Single voltage spikes demonstrate why local and global metrics must be reported together. Full-window ΔMAE remains below 4×10^{-4} SOC for every estimator and is effectively zero for DM, HDM, and DD. Event-aligned analysis, however, identifies a reproducible DD transient: its mean spike-sample penalty across cells is approximately 7.7×10^{-3} SOC, compared with near-zero direct penalties for the other representatives. Figures 12 and 13 further show that the DD response varies across cells and SOH states and then decays after the event. This pattern is consistent with a one-sample voltage impulse entering both the

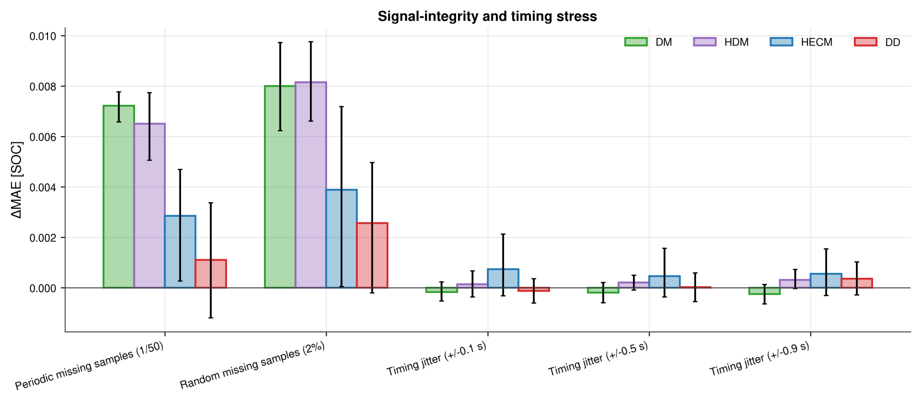

Figure 10: Six-cell signal-integrity and timing stress. Bars show cell-macro ΔMAE and hierarchical 95% confidence intervals for periodic and random sample loss and three timing-jitter levels. Sample loss clearly dominates timing jitter under the tested protocol.

absolute voltage and $\mathrm{d}U/\mathrm{d}t$ channels and perturbing the rolling recurrent prediction. It is a mechanism-supported interpretation, not proof of a unique causal input channel because no channel-removal ablation was performed.

4.7. Shared-SOH ablation

The primary comparison intentionally supplies one common causal LSTM-SOH trace to HDM, HECM, and DD. A paired ablation repeats selected scenarios with dataset reference SOH to quantify this shared-service confounder. Reference SOH reduces baseline MAE by 0.0370 for HDM and 0.0166 for HECM, but increases DD MAE by 0.0049 on average. The same direction persists for most noise, sample-loss, and dropout cases. At 3% current gain error, reference SOH still improves HDM and HECM but worsens DD by 0.0091. For temperature noise, the reference-minus-LSTM difference changes from its baseline value by only 0.0016 for HDM and less than 10^{-4} for HECM and DD, indicating that the incremental temperature-noise response is not dominated by the shared SOH service. This does not imply that a less accurate SOH estimate is intrinsically beneficial. Instead, the models consume SOH differently and were calibrated with different feature interactions. The LSTM-SOH primary results

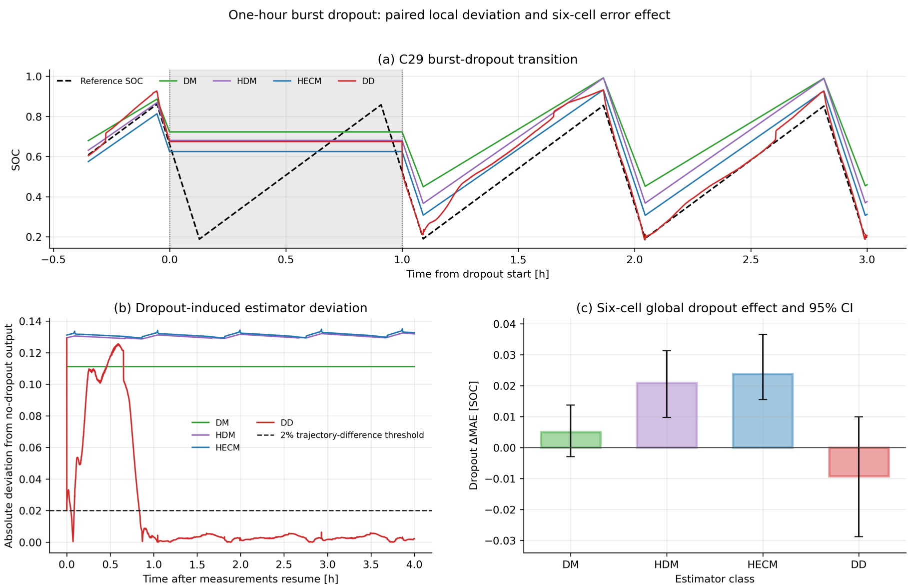

Figure 11: One-hour burst-dropout transition. (a) Estimator outputs around the shaded missing-input interval. (b) Absolute deviation from the paired duration-matched no-dropout output after measurements resume, with the 2% trajectory-difference line shown for context. (c) Six-cell global ΔMAE with hierarchical 95% confidence intervals.

therefore represent deployable pipeline performance, while reference-SOH runs are explanatory ablations rather than replacement rankings.

4.8. Robustness synthesis across representatives

Figure 14 condenses 17 measurement-corruption and signal-integrity rows and tests whether the nominal-accuracy ranking alone describes degraded-input behavior. It does not. Persistent gain error and missing information create large global penalties. Current-gain sensitivity increases with magnitude for every representative in the matched signed sweep, periodic and random sample loss particularly affect DM and HDM, and the one-hour dropout produces the largest point penalty for HECM. Timing jitter remains nearly neutral. Voltage spikes illustrate the opposite limitation of global averaging because their full-window MAE is small even though event alignment exposes a distinct DD transient. The eight-family aggregate shown later is only one weighting of these heteroge-

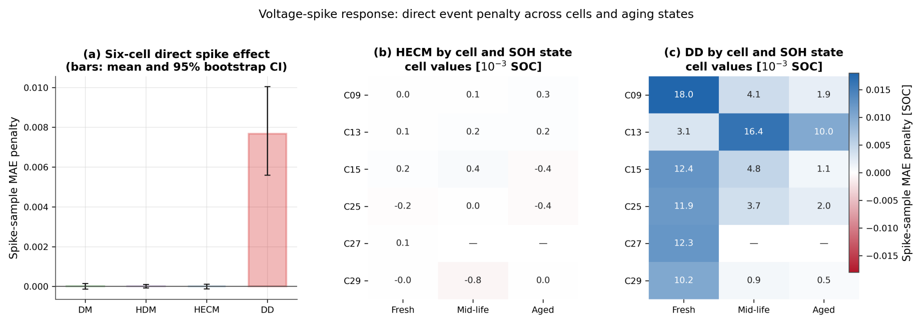

Figure 12: Voltage-spike sensitivity across six cells. (a) Mean event-sample MAE penalty with hierarchical 95% confidence intervals. (b,c) Cell- and SOH-state-resolved penalties for HECM and DD, respectively. Absent cell/state combinations are left blank. The event-aligned metric exposes the local DD response that is obscured in full-window ΔMAE.

neous effects. Its ranking is compared with alternative weightings rather than interpreted as an objective robustness winner. The central result is therefore a multi-dimensional and scenario-specific comparison rather than one universal ranking.

Under the family-balanced definition, the illustrative robustness scores are 0.562 for DM, 0.468 for HDM, 0.428 for HECM, and 0.842 for DD. Equal weighting per evaluated scenario yields 0.574, 0.504, 0.560, and 0.781, while the highest-level-plus-offset variant yields 0.615, 0.512, 0.576, and 0.665. DD remains the leading tested representative under these three aggregations, but its score and the ordering of HDM and HECM change materially. This exploratory synthesis supports DD for the evaluated disturbance mix without establishing an aggregation-independent ranking of estimator classes. The individual scenario results remain the primary robustness evidence.

4.9. ADC quantization

Figure 16 compares representative raw and quantized current and voltage signals and summarizes the resulting six-cell ΔMAE. The tested ADC steps ($\Delta I = 0.01\,\mathrm{A}$, $\Delta U = 0.005\,\mathrm{V}$, and $\Delta T = 0.5\,^{\circ}\mathrm{C}$) preserve the local signal shape but affect the tested representatives differently. DM and HDM increase by

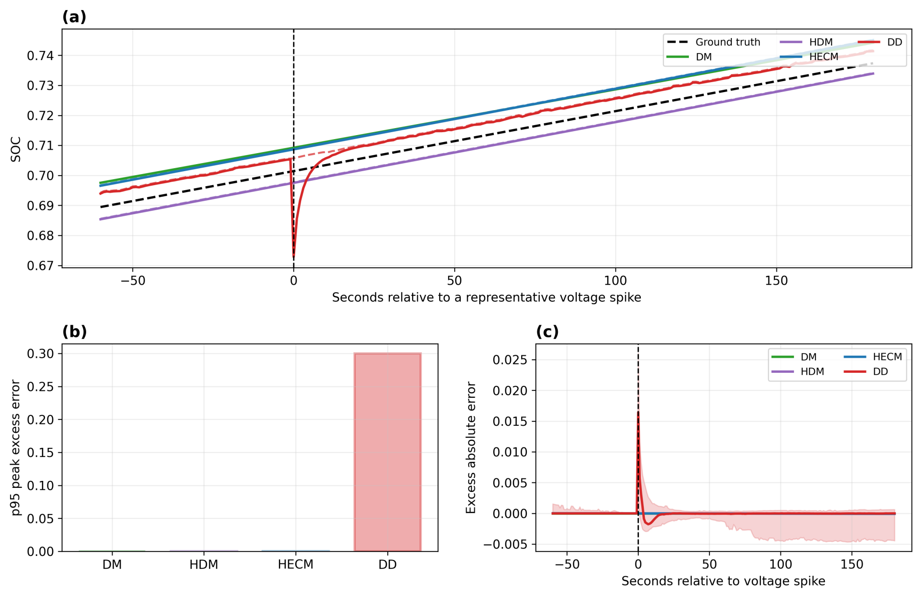

Figure 13: Event-aligned voltage-spike response. (a) Representative SOC trajectories around the injected spike, with the dashed black line showing dataset ground-truth SOC. (b) Model-wise 95th percentile peak excess error. (c) Mean excess absolute error aligned across repeated spikes, with the DD uncertainty envelope. The local response decays rapidly and is therefore weakly represented by full-window MAE.

0.0038 and 0.0035 MAE, respectively. HECM changes by -0.0018 and DD by $+0.0002$. All four confidence intervals overlap zero. Thus, quantization at these deliberately coarse steps is secondary to current gain error, sample loss, and burst dropout in this dataset, but it is not negligible for the integration-based models.

4.10. Embedded performance benchmark

The STM32 experiment first verifies numerical equivalence before comparing speed or memory. Across the six nominal cell sequences, mean hardware–software MAE remains below 4×10^{-4} percentage points for every SOC core and reaches 3.41×10^{-6} percentage points for DD (Figure 17). These differences are orders of magnitude below the estimator-to-dataset errors and show that the float32 C implementations reproduce their software references sufficiently closely

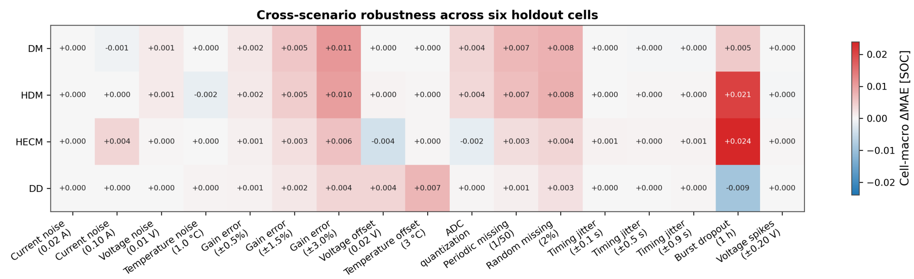

Figure 14: Cell-macro ΔMAE across 17 measurement-corruption and signal-integrity rows on six holdout cells. Current-gain rows use the adverse direction from the matched signed sweep in the final benchmark build. Positive values denote degradation relative to each model's paired baseline. Small negative values reflect finite-window signed effects and are not interpreted as beneficial disturbances. The initialization-mismatch response is evaluated separately in Figure 9.

for deployment measurements.

The isolated DM and HDM SOC updates both have sub-microsecond median latency (0.171 and 0.165 μs), HECM requires 23.9 μs, and continuous-state DD requires 424.9 μs (Figure 18). At the nominal 1-Hz update rate, these medians occupy less than 0.043% of one sampling period, although this isolated deadline share is not a measurement of total BMS CPU load. The DD value in this cross-model comparison is the continuous-state deployment variant. Exact 2024-sample rolling-window semantics are evaluated separately because recomputing the full sequence at every sample is much slower.

For the cross-model embedded performance comparison, DD denotes the continuous-state deployment variant used in the latency analysis. The compiled DM, HDM, HECM, and DD firmware occupies 27.0, 27.1, 470.1, and 115.9 KiB of flash, respectively. The corresponding measured peak runtime RAM from the representative C09 replay is 2.4, 2.4, 2.5, and 4.1 KiB (Figure 19). In bytes, DM and HDM each combine 872 B of statically allocated variable storage with a 320 B maximum dynamic-memory allocation and 1240 B of maximum call-stack use, yielding 2432 B. HECM reaches 2528 B because its static variable allocation increases to 968 B. Continuous-state DD combines 2588 B of static variable

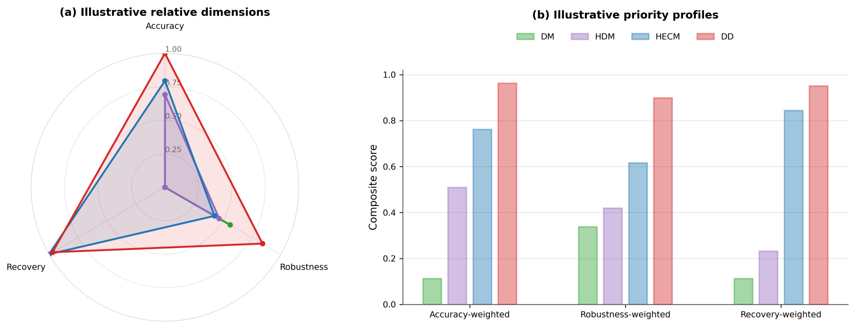

Figure 15: Decision-oriented synthesis for the four tested representatives. (a) Relative min–max-normalized accuracy, family-balanced robustness across eight evaluated disturbance families, and observed persistent paired recovery. The current-gain component uses the adverse direction from the matched signed sweep. (b) Composite scores under three illustrative 60/20/20 priority profiles. Scores summarize the present benchmark only, depend on the selected weighting and intervention conventions, and do not replace scenario-level evidence.

storage with 320 B of maximum dynamic-memory allocation and 1248 B of maximum call-stack use, yielding 4156 B. HECM flash occupancy is dominated by its parameter surfaces, whereas the DD parameter storage accounts for most of its flash demand. These firmware-level values do not include the shared SOH-LSTM service or concurrent BMS tasks.

Figure 20 makes the DD deployment trade-off explicit, while Figure A.24 provides the corresponding latency distributions over all six hardware cell sequences. Exact rolling-window inference preserves the trained software semantics but requires roughly 724 ms median inference time, or 72.4% of the one-second period, and 67.3 KiB peak runtime RAM. Continuous-state inference reduces latency to approximately 0.425 ms and peak RAM to 4.1 KiB while retaining similar error on the six nominal sequences. Periodically resetting that state has a comparable 4.1 KiB peak but creates much larger transient errors, with a maximum SOC error near 28 percentage points. Continuous state is therefore the practical hardware candidate, but it remains an inference-semantics abla-

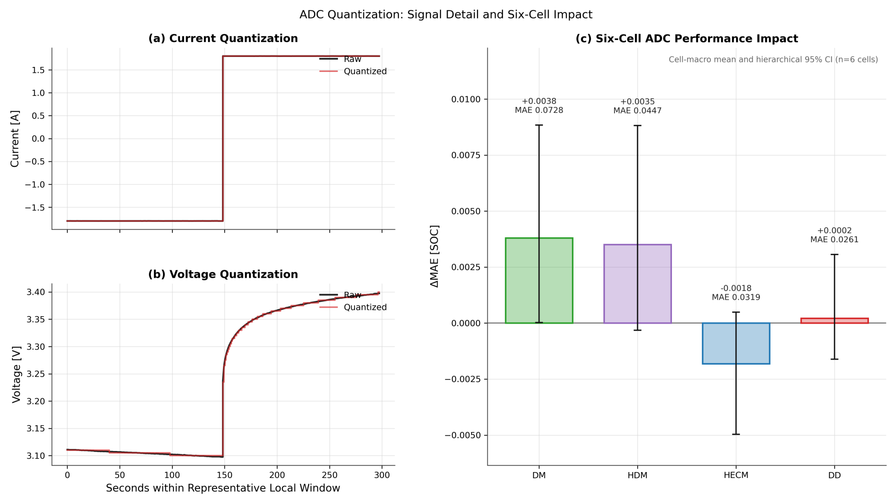

Figure 16: ADC quantization. (a,b) Representative raw and quantized current and voltage. (c) Six-cell ΔMAE with hierarchical 95% confidence intervals and resulting quantized-signal MAE annotations.

tion rather than a drop-in replacement for the primary rolling-window software benchmark.

5. Discussion

5.1. Structural interpretation of the representatives

Because one concrete implementation represents each broad estimator class, the results support mechanism-level interpretation but not family-wide performance claims. DM behaves as expected for a current-integration representative because it has the largest nominal error and is sensitive to persistent gain error and missing samples when no voltage-feedback correction is available. Its practical full-charge anchor limits unbounded drift, but the lifecycle experiment shows that the reset does not erase the accumulated gain-error contribution uniformly. HDM adapts the capacity denominator through learned SOH and can reduce capacity mismatch, yet it retains the same charge-integration core. Its similar response to current-gain error and missing charge information is therefore structural to this implementation rather than removed by SOH adaptation.

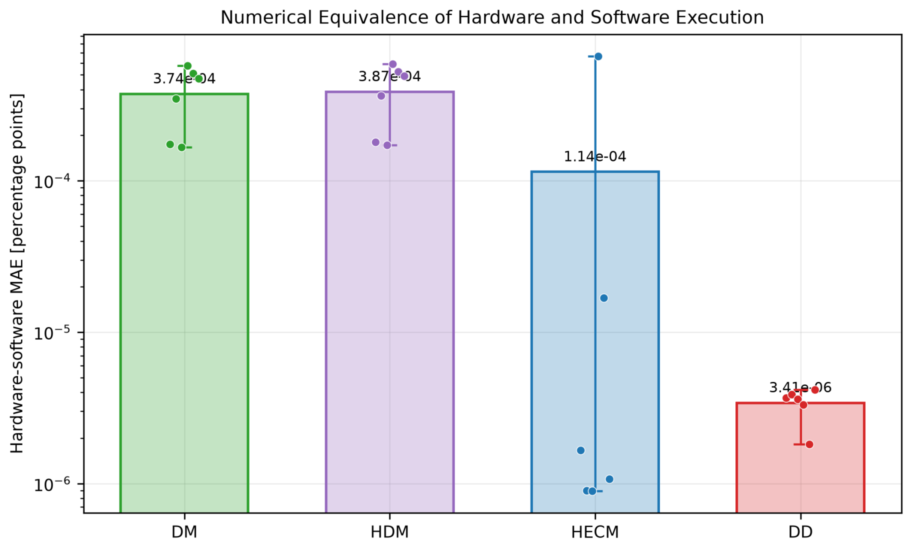

Figure 17: Numerical equivalence of STM32 and software execution over six cell sequences. Bars show mean hardware–software MAE in percentage points, dots show cells, and whiskers show the cell range. The logarithmic scale emphasizes that all implementation differences are negligible relative to model error.

HECM combines current-driven state propagation with a voltage residual and has the strongest observed recovery profile under the tested initialization mapping. Returns before the common 0.562-h observation start remain unresolved. This correction capability does not make it universally robust. High current noise and burst dropout produce the largest HECM point penalties because both state propagation and observer correction are affected when the causal input stream is corrupted or frozen, although the high-current-noise confidence interval spans zero. The dedicated lookup analysis shows that the lower HECM current-gain penalty is retained under systematic $\pm 10\%$ resistance scaling and $\pm 10\,\mathrm{mV}$ OCV shifts. Its largest absolute macro interaction is 0.00026 MAE. The same perturbations shift baseline accuracy more visibly, so precise lookup calibration remains relevant even though the tested current-gain response is locally stable. Its table dependence and flash footprint must therefore be balanced against its observed initialization correction.

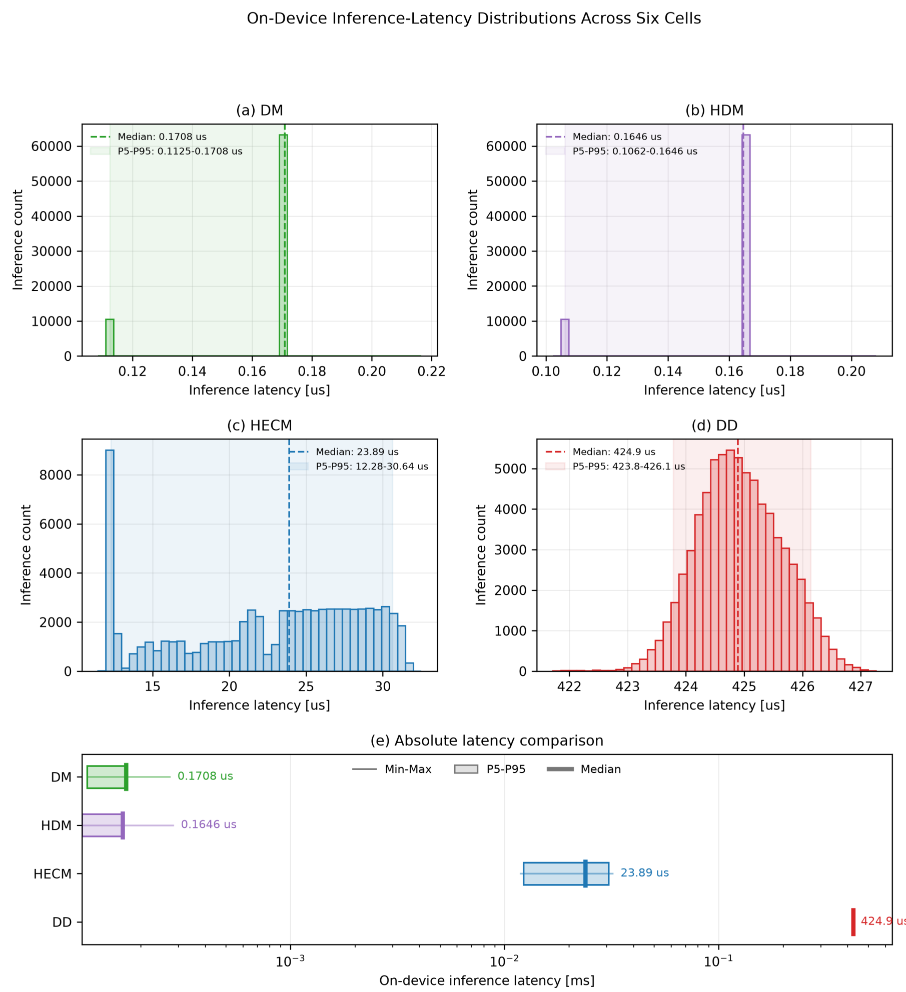

Figure 18: On-device SOC-core latency distributions across six cells and three replay rounds. Panels (a–d) show per-inference distributions. Panel (e) compares minimum, 5th percentile, median, 95th percentile, and maximum on a logarithmic scale. DD denotes continuous-state inference in this cross-model latency comparison.

DD has the lowest nominal cell-macro error and the best adverse current-gain score in this comparison. It is also comparatively resilient to isolated sample loss. Conversely, event-aligned voltage spikes expose a recurrent transient that full-window MAE hides. This vulnerability is consistent with the selected voltage and derivative features and rolling recurrent context. It is specific to the

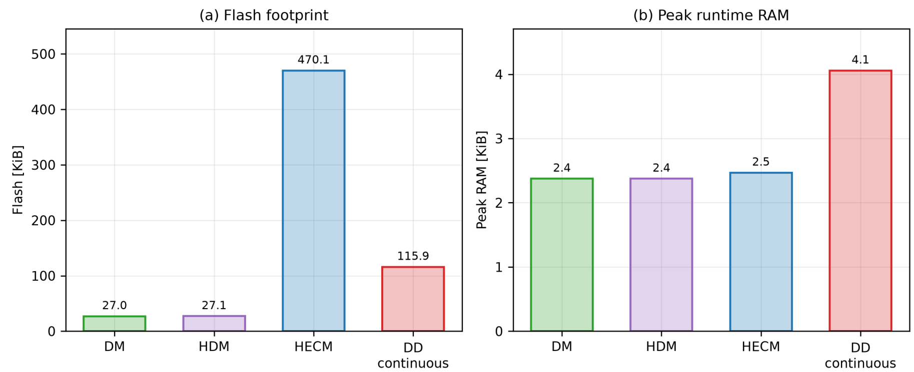

Figure 19: Embedded memory requirements of the isolated STM32 SOC implementations. (a) Compiled flash occupancy. (b) Peak runtime RAM measured during a representative 2025-sample C09 replay, combining statically allocated variable storage with maximum dynamic-memory allocation and call-stack use. DD denotes continuous-state execution, matching Figure 18. Exact rolling-window execution is compared separately in Figure 20.

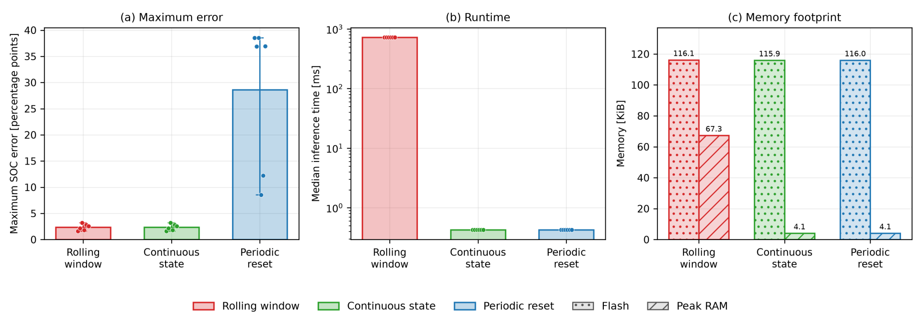

Figure 20: DD inference-mode trade-off on the STM32. The panels compare maximum dataset-SOC error, median latency, compiled flash occupancy, and measured peak runtime RAM for exact rolling-window, continuous-state, and periodically reset recurrent execution. Cell points show that periodic resets, not continuous forwarding, create the dominant accuracy loss.

frozen GRU, feature set, training data, and inference semantics, not a universal limitation of recurrent or data-driven SOC estimation.

5.2. Why nominal accuracy is insufficient

Nominal MAE alone fails to reveal the conditions under which the clean-data ranking changes or becomes incomplete. It does not expose the DD voltage-spike transient, the persistence of HECM after a first recovery hold, the shared-SOH feature coupling, or the orders-of-magnitude difference in execution cost. It also cannot resolve how alternative weights across heterogeneous disturbance families change a normalized robustness score. No single scalar captures these deployment trade-offs. Selection should therefore begin with the relevant disturbance and recovery requirements and then consider latency, memory, and state-service architecture rather than treating baseline accuracy as a sufficient proxy.

5.3. Implications for BMS deployment

Among the tested representatives, DD has the lowest nominal error against the dataset ground truth and remains a strong candidate when its recurrent implementation cost and voltage-spike response are acceptable. HECM is preferable when the observed persistent correction after the tested state mismatch dominates and a larger flash image is available. DM and HDM remain attractive when deterministic sub-microsecond execution and small memory dominate, but persistent current calibration, reliable sampling, and full-charge anchoring then become stronger system-level requirements. These are implementation-level recommendations, not claims that one estimator family is intrinsically superior.

The STM32 results narrow the gap between algorithmic and deployment claims. All four isolated SOC cores meet the one-second deadline in the tested replay, but their margins differ by more than six orders of magnitude between DM/HDM and exact rolling-window DD. Continuous-state DD removes most of that cost, at the price of changing inference semantics. The hardware test includes firmware-level measurements of maximum call-stack and dynamic-memory use but does not include the shared SOH-LSTM, energy, pack communication, or concurrent BMS tasks. It therefore supports isolated SOC-core execution under the tested replay, not complete BMS feasibility or schedulability. Table 3 summarizes the evidence without hiding these boundaries.

Figure 15 complements the recommendation map with relative scores, not additional measurements. Lower-is-better inputs are min–max normalized within the four tested implementations, and non-positive disturbance changes contribute zero robustness penalty. Accuracy combines baseline MAE, RMSE, and 95th-percentile error. Robustness gives equal weight to eight evaluated disturbance families after averaging levels within each family, with the current-gain family taken from the adverse direction of the matched signed sweep. Recovery uses observed persistent recovery-or-censor time, excess-error area, and persistent censoring from the canonical paired initialization analysis. The three profiles assign 60% to their named dimension and 20% to each remaining dimension. A separate sensitivity analysis compares this family-balanced definition with equal weighting per scenario and a high-severity variant. The resulting scores are illustrative because their ordering depends on the selected aggregation rule and recovery intervention convention. They neither remove scenario-specific weaknesses nor support extrapolation from these representatives to their complete model families.

5.4. Limitations and future extensions

Only one frozen implementation represents each broad estimator family. Different reset logic, ECM order and parameterization, neural architecture, input features, training data, or inference semantics can change the ranking. The study is further limited to six holdout cells from one controlled LFP dataset and therefore cannot establish transfer to NMC/NCA chemistry, other form factors, pack imbalance, other thermal environments, or field duty cycles. The load groups are descriptive strata with $n = 2$ low-load, $n = 3$ middle-load, and one exploratory high-load cell. They are not randomized treatments. C27 contains no mid-life or aged window, and missing states are not imputed. With six independent cells, confidence intervals and effect magnitudes are more informative than dichotomized significance decisions. The dataset SOC ground truth is

Table 3: Decision-oriented recommendation map derived from the benchmark results. The table identifies the strongest observed representative under one dominant deployment priority and the stated protocol boundaries.

Primary deployment priority	Selected representative	Main supporting benchmark evidence	Main trade-off to consider
Lowest nominal SOC error	DD	Lowest six-cell macro MAE (0.0258) and RMSE (0.0300)	Rolling-window execution is the most computationally expensive implementation
Strongest observed recovery from initialization mismatch	HECM	Lowest observed persistent-recovery-or-censor time and substantially fewer first-entry relapses than DD	Model-specific interventions and left-censoring before 0.562 h limit an exact time ranking
Lowest adverse current-gain penalty	DD	Matched signed sensitivity gives ΔMAE 0.0038 at 3%, below HECM, HDM, and DM	Temperature-offset and local voltage-spike sensitivity remain visible
Smallest observed periodic/random sample-loss penalty	DD	ΔMAE 0.0011/0.0026	The duration-matched dropout effect is non-positive but highly variable across cells
Lowest deterministic compute cost	DM/HDM	Sub-microsecond median SOC-core latency and approximately 27 KiB flash	Integration structure is sensitive to gain error and missing charge information
Practical fast neural deployment	DD continuous state	Approximately 0.425 ms median and 4.1 KiB peak runtime RAM	Different recurrent semantics from the primary rolling-window benchmark

reconstructed offline by Coulomb counting and voltage-anchor logic. Reported nominal rankings therefore apply to this consistently used ground-truth construction.

The primary cross-model campaign manipulates only measurement, timing, availability, and initialization channels. The separate HECM lookup analysis covers uniform local resistance and OCV deviations but not arbitrary table-shape error, SOH-axis error, thermal parameter drift, or combined parameter mismatch. Cell/pack parameter mismatch outside these probes, ECM aging drift, hysteresis error, thermal gradients, sensor cross-sensitivity, balancing and contactor faults, communication-protocol failures, numerical faults, and deadline interference remain outside the study. DD initialization is implemented through an SOC-equivalent Q_c offset rather than direct hidden-state corruption. Recovery observation starts 0.562 h after the intervention, so earlier endpoints are left-censored and the estimator-specific interventions do not support an exact latent-state convergence ranking. The signed current-gain sweep evaluates symmetric sensor-gain errors, while the continuous C29 replay provides a single-cell lifecycle diagnostic rather than six-cell lifecycle evidence. Abrupt capacity-loss events are absent, so the hourly SOH service is evaluated only for gradual aging. The voltage-spike mechanism is supported by event alignment but not proven by an input-channel ablation. The HPPC identification records and fitted surfaces remain required release artifacts even though their development pool excludes the holdout cells.

The hardware experiment isolates SOC cores and passes SOH as an external causal trace. Its runtime-RAM watermark comes from a representative C09 replay, not every cell, and the exact rolling DD stack maximum is based on two valid post-warm-up calls in that replay. The profile covers the benchmark firmware but not an end-to-end SOH service or concurrent BMS tasks. Energy, total scheduled CPU utilization, and pack-level closed-loop behavior also remain outside the experiment. Future work should therefore extend memory and energy profiling to the combined SOC–SOH service, add scheduling tests under concurrent BMS workloads, and investigate clustered faults, hidden-state behavior, additional chemistries, and further duty cycles.

6. Conclusion

This study benchmarked the robustness of representatives from four SOC-estimation classes under disturbed sensing and embedded constraints. Nominal accuracy against the dataset ground truth alone is insufficient for estimator selection. DD provides the lowest nominal error, but its local voltage-spike response, initialization relapses, and rolling-window cost are invisible in a clean-data MAE ranking. HECM provides the strongest observed recovery profile under the tested model-specific initial mismatch, but has the largest point penalties under high current noise and prolonged dropout and requires the largest flash image. Its lower current-gain penalty remains stable under the tested local resistance and OCV lookup perturbations. The current-noise interval spans zero, and recovery before the common observation start remains unresolved. DM and

HDM execute with minimal latency and memory, yet remain most dependent on accurate, continuous current integration and physical re-anchoring. Aggregate robustness scores are weighting-dependent and include the adverse direction from the matched signed current-gain sweep. They therefore remain illustrative rather than defining a universal winner.

The practical outcome is therefore a selection framework rather than a universal class ranking. Accuracy characterizes agreement with the dataset ground truth, robustness identifies disturbance-specific degradation, recovery describes observable re-synchronization under stated interventions, and hardware measurements test isolated execution cost. The STM32 tests demonstrate one-second execution of all isolated SOC cores in the tested replay while exposing substantial implementation trade-offs. They do not establish full-BMS feasibility. These conclusions apply to the selected Coulomb-counting, SOH-adaptive, ECM–EKF, and GRU representatives on the tested LFP aging data. Broader claims require additional implementations, independently documented model provenance and SOC ground truths, chemistries, operating profiles, and complete SOC–SOH hardware scheduling.

Appendix A. Holdout Coverage and Frozen Protocol

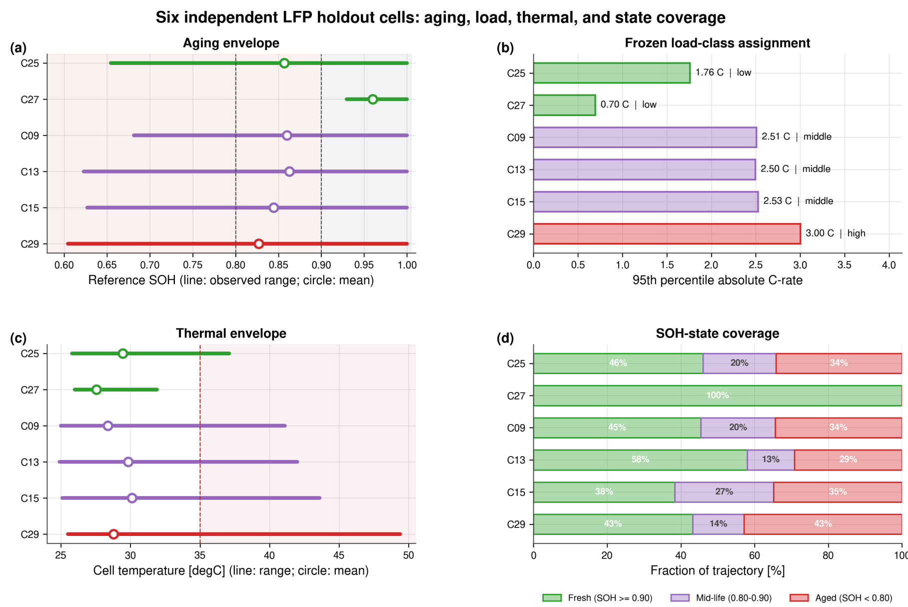

Figure A.21: Quantitative coverage of the six independent holdout cells. Panels show observed SOH, the descriptive load-class assignment, thermal envelope, and full-trajectory SOH-state coverage. High-load evidence consists only of C29 and is exploratory.

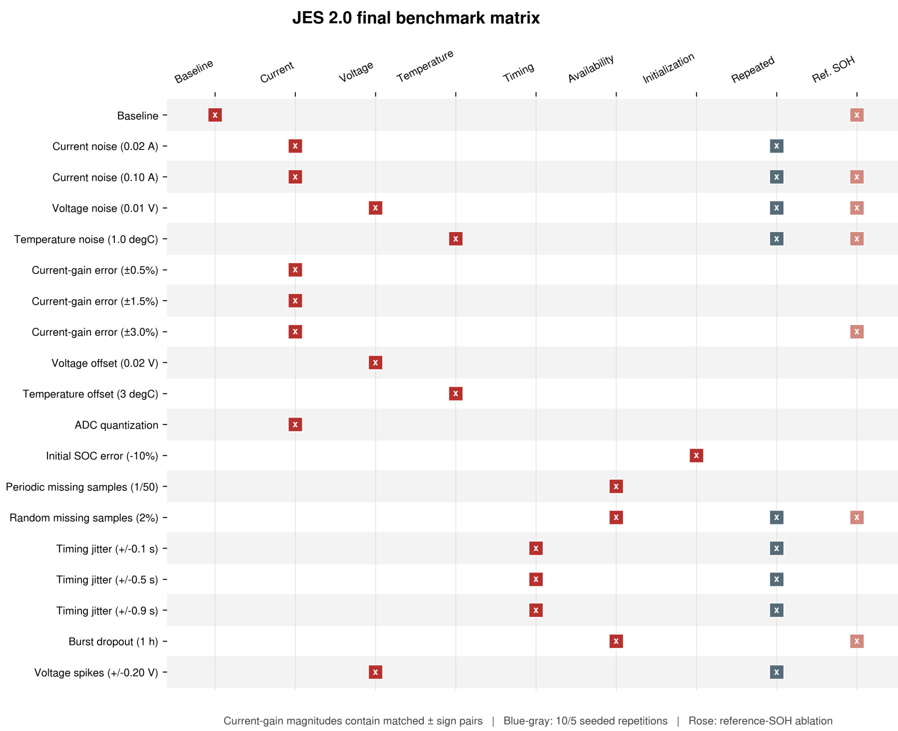

Figure A.22: Complete 19-scenario final JES2 matrix. Red markers identify the manipulated measurement, timing, availability, or initialization channel. Blue-gray markers denote repeated stochastic cases. Rose markers identify paired reference-SOH ablations. Each current-gain magnitude contains matched positive and negative sign sublevels.

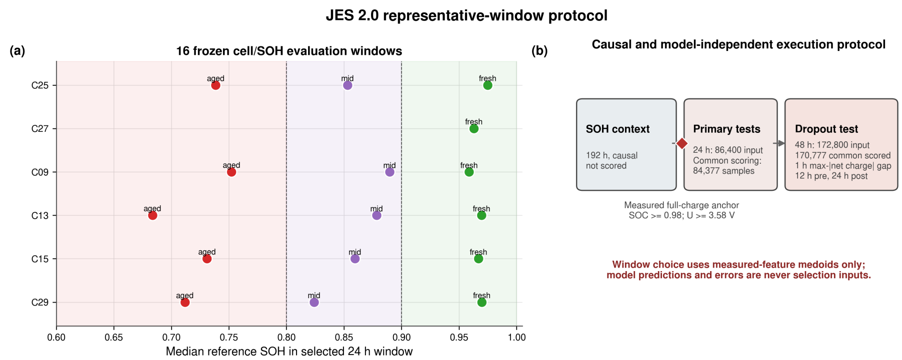

Figure A.23: Frozen representative-window protocol. Sixteen cell/SOH windows are selected using measured-feature medoids only. Every estimator is scored from source sample 2023, yielding 84,377 matched points per 24-h window after a 192-h causal SOH context. The dropout case uses 48 h, places the gap at the eligible one-hour interval with maximum absolute net charge, and contributes 170,777 matched points.

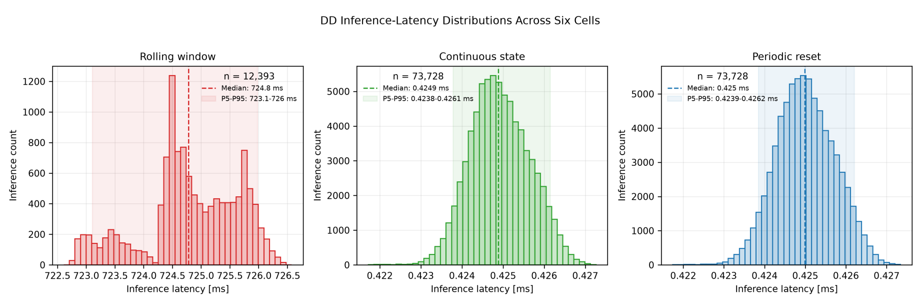

Figure A.24: DD latency distributions over the six hardware cell sequences. Exact rolling-window execution is shown separately from continuous-state and periodically reset execution because their scales and recurrent semantics differ substantially.

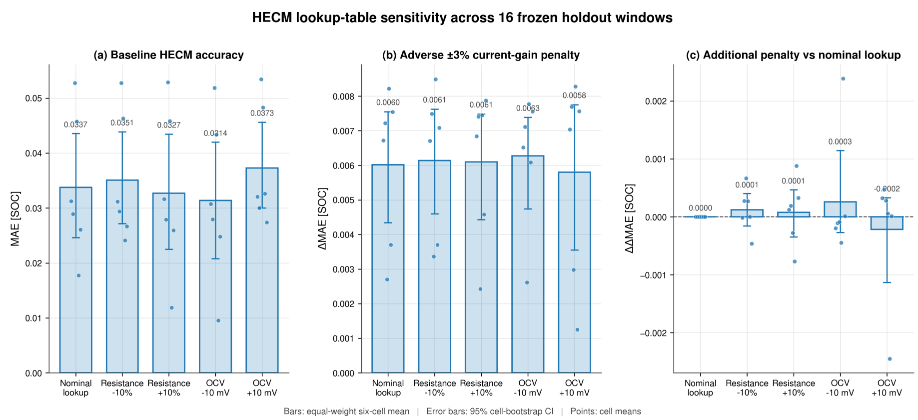

Figure A.25: HECM lookup-table sensitivity over all 16 frozen windows. (a) Baseline MAE under the nominal lookup, joint ±10% scaling of R_i, R_1, and R_2, and uniform $\pm 10\,\mathrm{mV}$ OCV shifts. (b) Adverse current-gain penalty from matched -3% and $+3\%$ gain errors relative to the baseline under the same lookup condition. (c) Lookup–gain interaction relative to the nominal-table penalty. Bars show equal-weight six-cell means, error bars show 95% cell-bootstrap intervals, and points show cell means.

Appendix B. Supplementary Numerical Results

Table B.4: Six-cell baseline cell-macro statistics with hierarchical 95% confidence intervals.

Estimator	MAE [SOC]	RMSE [SOC]	P95 error [SOC]
DM	0.0690 [0.0499, 0.0823]	0.0740 [0.0532, 0.0881]	0.1088 [0.0768, 0.1307]
HDM	0.0412 [0.0301, 0.0515]	0.0442 [0.0319, 0.0557]	0.0649 [0.0449, 0.0846]
HECM	0.0337 [0.0246, 0.0434]	0.0388 [0.0286, 0.0493]	0.0644 [0.0468, 0.0821]
DD	0.0258 [0.0189, 0.0325]	0.0300 [0.0223, 0.0366]	0.0508 [0.0375, 0.0619]

Table B.5: Exact paired baseline-MAE comparisons across the six holdout cells. Differences are defined as estimator B minus estimator A. Holm correction is applied across the six model-pair tests.

Estimator A	Estimator B	Mean difference [SOC]	95% CI	d_z	p_{exact} / p_{Holm}
DM	HDM	-0.0278	[-0.0381, -0.0171]	-1.93	0.0313 / 0.1875
DM	HECM	-0.0352	[-0.0470, -0.0224]	-2.00	0.0313 / 0.1875
DM	DD	-0.0431	[-0.0558, -0.0286]	-2.29	0.0313 / 0.1875
HDM	HECM	-0.0075	[-0.0162, 0.0015]	-0.61	0.1563 / 0.1875
HDM	DD	-0.0154	[-0.0243, -0.0084]	-1.39	0.0313 / 0.1875
HECM	DD	-0.0079	[-0.0176, -0.0005]	-0.68	0.0938 / 0.1875

Table B.6: Selected cell-macro scenario penalties relative to paired baseline. Current-gain values use the adverse sign from the matched ±3% sweep in the final benchmark build.

Scenario	DM	HDM	HECM	DD
Current noise, $\sigma_I = 0.10\,\mathrm{A}$	-0.0009	+0.0000	+0.0038	-0.0000
Current gain, adverse 3%	+0.0108	+0.0101	+0.0060	+0.0038
Voltage offset, +0.02 V	+0.0000	+0.0005	-0.0040	+0.0038
Temperature offset, +3 °C	+0.0000	+0.0002	+0.0002	+0.0074
ADC quantization	+0.0038	+0.0035	-0.0018	+0.0002
Random missing samples, 2%	+0.0080	+0.0081	+0.0039	+0.0026
Timing jitter, ±0.9 s	-0.0003	+0.0003	+0.0006	+0.0003
Burst dropout, 1 h	+0.0050	+0.0209	+0.0239	-0.0093
Voltage spikes, ±0.20 V	+0.0000	-0.0003	+0.0002	-0.0001

Table B.7: Sensitivity of the illustrative normalized robustness score to the aggregation rule.

Estimator	Family balanced	Equal scenario	Highest level plus offsets
DM	0.562	0.574	0.615
HDM	0.468	0.504	0.512
HECM	0.428	0.560	0.576
DD	0.842	0.781	0.665

Table B.8: Canonical paired initialization recovery and target-hardware summary. Recovery values are equal-weight cell means on the common source-sample interval, which begins 0.562 h after the intervention. Endpoints already satisfied at this boundary are recorded at the boundary as conservative upper bounds.

Metric	DM	HDM	HECM	DD
First 300-s entry/censor time [h]	22.28	17.21	1.12	1.29
Persistent recovery/censor time [h]	22.28	22.28	1.20	2.60
Recovery excess-error AUC [SOC h]	1.735	1.738	0.078	0.116
First-entry censored fraction	0.89	0.67	0.00	0.00
Persistent-recovery censored fraction	0.89	0.89	0.00	0.00
Relapse fraction after first hold	0.00	0.22	0.17	0.78
Median SOC-core latency [μs]	0.171	0.165	23.9	424.9[a]
Flash [KiB]	27.0	27.1	470.1	115.9[a]
Peak runtime RAM [KiB]	2.4	2.4	2.5	4.1[a]

[a]Continuous-state DD. Exact rolling-window execution requires approximately 724 ms and 67.3 KiB peak RAM.

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CRediT authorship contribution statement

The author contribution statement is provided in the non-blinded submission materials.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT, DeepL, and Grammarly in order to improve the linguistic quality and style of the manuscript. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Data availability

The dataset is publicly available through its cited TU Berlin repository record [28]. Code and results are maintained in a public GitHub repository. Before resubmission, a versioned release will archive the fixed experimental definitions and random seeds, trained model parameters, input scalers, ECM parameterization with originating-test provenance, run summaries, raw hardware summaries and timing records, exact clean firmware revision, analysis and plotting procedures, and software environment needed to reproduce all reported results.

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